Physics Lectures

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.

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Contents

Preface: What These Notes Are For

Remark

These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether's theorem and conserved currents,
- Green's functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

$$\text{classical field} \longrightarrow \text{mode expansion} \longrightarrow \text{operators} \longrightarrow \text{states} \longrightarrow \text{correlators} \longrightarrow \text{amplitudes.}$$

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

A free field is the continuum limit of a system with many coupled degrees of freedom. Its normal modes behave like independent harmonic oscillators. The first idea of quantum field theory is to quantize these oscillators and to interpret their excitations as field quanta.

1.1 The Purpose of the First Step

The transition from classical field theory to quantum field theory can look too large if one begins immediately with operator-valued distributions, Fock spaces, and relativistic commutators. This chapter takes a slower route. It explains why the harmonic oscillator algebra appears at the beginning of quantum field theory.

The guiding chain of ideas is

$$\text{classical field} \longrightarrow \text{normal modes} \longrightarrow \text{classical oscillators} \longrightarrow \text{quantum oscillators} \longrightarrow \text{field quanta}.$$

The goal is not yet to construct the full canonical field algebra. That will be done in the next chapter. Here we only identify the mechanical structure hidden inside a free field.

1.2 A Finite System of Coupled Oscillators

Consider a finite chain of N identical masses connected by springs. Let $q_j(t)$ be the displacement of the j-th mass from equilibrium. A typical quadratic Lagrangian has the form

$$L = \sum_{j=1}^{N} \frac{m}{2}\dot{q}_j^2 - \sum_{j=1}^{N} \frac{\kappa}{2}(q_{j+1} - q_j)^2,$$

with boundary conditions chosen so that the notation is meaningful. The coordinates q_j are coupled because the potential energy depends on differences between neighbouring displacements.

The normal-mode method replaces the original coordinates by new coordinates $Q_r(t)$,

$$q_j(t) = \sum_r Q_r(t) u_r(j),$$

where $u_r(j)$ are spatial mode functions. With an appropriate normalization, the Lagrangian becomes diagonal:

$$L = \sum_r \left(\frac{1}{2}\dot{Q}_r^2 - \frac{1}{2}\omega_r^2 Q_r^2 \right).$$

Thus the coupled system is equivalent to a collection of independent harmonic oscillators.

Example

The original variables q_j describe local displacements of the masses. The normal coordinates Q_r describe collective oscillation patterns of the whole chain. Quantization is easiest in the normal coordinates because the modes are independent.

1.3 The Field as a Continuum Limit

A field is the continuum analogue of a system with one degree of freedom at each spatial point. In a one-dimensional chain with lattice spacing a, the site position is $x_j = ja$, and one may think schematically of

$$q_j(t) \approx \phi(t, x_j).$$

As $a \to 0$, sums over sites become spatial integrals, and discrete normal modes become Fourier modes.

For a real scalar field in a finite box of volume V, periodic boundary conditions make the allowed momenta discrete. A schematic Fourier expansion is

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}}.$$

For a real field the complex coefficients are not all independent; one must impose the appropriate reality condition. Equivalently, one may use real sine and cosine modes. The important point is independent of this choice: after the free field is decomposed into normal modes, each independent mode behaves like a classical oscillator.

1.4 Each Free Mode Is a Harmonic Oscillator

The free real scalar field has Lagrangian

$$L = \int \mathrm{d}^3x \left(\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 \right).$$

Its equation of motion is the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

After the free field is diagonalized into normal modes, the Lagrangian takes the schematic oscillator form

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2}\dot{q}_{\mathbf{k}}^2 - \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The canonical momentum of a mode is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian is

$$H = \sum_{\mathbf{k}} H_{\mathbf{k}}, \qquad H_{\mathbf{k}} = \frac{1}{2}p_{\mathbf{k}}^2 + \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2.$$

A free scalar field is therefore a collection of independent harmonic oscillators, one for each independent normal mode.

Remark

Interactions destroy this simple diagonal form. Perturbative QFT begins by quantizing the free oscillators exactly and then treating interactions as corrections to the free theory.

1.5 Quantizing One Harmonic Oscillator

Before quantizing all field modes, recall the single harmonic oscillator. The classical variables q and p become operators $\hat{q}$ and $\hat{p}$ satisfying

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

Introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega}{2}}\,\hat{q} + \frac{i}{\sqrt{2\omega}}\,\hat{p}, \qquad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\,\hat{q} - \frac{i}{\sqrt{2\omega}}\,\hat{p}.$$

Then

$$[\hat{a}, \hat{a}^\dagger] = 1,$$

and

$$\hat{H} = \omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

The operator $\hat{a}^\dagger\hat{a}$ counts oscillator excitations.

1.6 Number States and the Oscillator Vacuum

The oscillator vacuum is defined by

$$\hat{a}|0\rangle = 0.$$

The excited states are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \qquad n = 0, 1, 2, \ldots.$$

If

$$\hat{N} = \hat{a}^\dagger\hat{a},$$

then

$$\hat{N}|n\rangle = n|n\rangle, \qquad \hat{H}|n\rangle = \omega\left(n + \frac{1}{2}\right)|n\rangle.$$

The vacuum is not a state of zero energy. It has the zero-point energy $\omega/2$. This fact becomes more delicate for fields because a field has infinitely many oscillator modes.

1.7 From One Oscillator to Many Field Modes

For a free field, every independent mode is quantized as an oscillator. Thus one introduces a pair of operators for each mode,

$$\hat{a}_{\mathbf{k}}, \qquad \hat{a}^\dagger_{\mathbf{k}}.$$

In a finite box their algebra is

$$[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = \delta_{\mathbf{k},\mathbf{p}}, \qquad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \qquad [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = 0.$$

The free-field vacuum is the state annihilated by all annihilation operators,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for all } \mathbf{k}.$$

A one-quantum state is obtained by applying one creation operator:

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

A many-quantum state is obtained by applying several creation operators.

Definition

The *Fock space* of the free field is the Hilbert space generated from the vacuum by applying creation operators for all modes. It naturally contains states with zero, one, two, and more field quanta.

1.8 Field Quanta as Mode Excitations

For the free theory, the Hamiltonian has the schematic form

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The operator $\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}$ counts how many quanta occupy the mode $\mathbf{k}$. The state $\hat{a}^\dagger_{\mathbf{k}}|0\rangle$ has one quantum of energy $\omega_{\mathbf{k}}$ above the free vacuum and momentum $\mathbf{k}$. In the free theory, this is what one means by a one-particle state.

This interpretation is precise for a free field. In interacting QFT, particles can be dressed by interactions, unstable particles appear as resonances, and the vacuum itself can be nontrivial. Nevertheless, the free Fock-space picture is the starting point for perturbation theory and scattering.

1.9 What Is Still Missing?

The oscillator construction explains why creation and annihilation operators appear, but it is not yet the full canonical quantization of a field. We have not yet imposed commutation relations directly on $\hat{\phi}(t, \mathbf{x})$ and its conjugate momentum $\hat{\pi}(t, \mathbf{x})$. We have also not fixed the continuum normalization of the mode expansion, nor have we discussed fields as operator-valued distributions.

These are the tasks of the next chapter. Chapter 2 turns the oscillator picture into the canonical quantization of the real scalar field.

Summary

A free classical field can be diagonalized into independent normal modes. Each mode is a harmonic oscillator. Quantizing those oscillators gives creation and annihilation operators. Acting with creation operators on the vacuum builds Fock space, and the resulting mode excitations are interpreted as free particles. This oscillator picture is the conceptual bridge to canonical field quantization.

1.10 Chapter Checklist

After reading this chapter, one should be able to:

- explain why a free field decomposes into normal modes;
- explain why each independent mode behaves as a harmonic oscillator;
- quantize a single harmonic oscillator using $\hat{a}$ and $\hat{a}^\dagger$;
- define the oscillator vacuum and number states;
- describe the free-field vacuum and one-particle states;
- explain why free particles are mode excitations;
- state what remains to be done in canonical field quantization.

Chapter 2

Canonical Quantization of the Real Scalar Field

Remark

Chapter 1 explained why a free field should be thought of as a collection of oscillators. This chapter makes that statement precise. We impose canonical commutation relations on the field and its conjugate momentum, derive the standard mode expansion, and recover the Fock-space interpretation from the field algebra itself.

2.1 From the Oscillator Picture to Field Operators

The previous chapter started from normal modes. It argued that a free scalar field is a collection of harmonic oscillators labelled by momentum. That is the right intuition, but it is not yet the full canonical quantization of a field.

The full canonical construction begins before the normal-mode expansion. It starts with the classical field $\phi(t, \mathbf{x})$, its conjugate momentum $\pi(t, \mathbf{x})$, and their Poisson brackets. Quantization then promotes ϕ and π to operators and replaces the Poisson brackets by commutators.

The logic of this chapter is therefore

$$\text{classical field variables} \longrightarrow \text{equal-time commutators} \longrightarrow \text{operator mode expansion} \longrightarrow \text{creation and annihilatio}$$

This is the formal version of the oscillator picture developed in Chapter 1.

2.2 Classical Starting Point

We use units $c = \hbar = 1$ and the metric convention $(+, -, -, -)$. The free real scalar field is described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2.$$

The conjugate momentum is

$$\pi(t, \mathbf{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}).$$

The Euler–Lagrange equation is the Klein–Gordon equation,

$$(\Box + m^2)\phi(x) = 0,$$

where $x = (t, \mathbf{x})$. In components this reads

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

The classical Hamiltonian density is

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2,$$

so the Hamiltonian is

$$H = \int d^3x \left(\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2\right).$$

This is the continuum field-theoretic analogue of the oscillator Hamiltonian $p^2/2 + \omega^2 q^2/2$.

2.3 Equal-Time Commutation Relations

The classical canonical Poisson bracket is

$$\{\phi(t,\mathbf{x}),\pi(t,\mathbf{y})\}_{\mathrm{PB}}=\delta^{(3)}(\mathbf{x}-\mathbf{y}).$$

Canonical quantization promotes ϕ and π to operators $\hat{\phi}$ and $\hat{\pi}$, and imposes

$$[\hat{\phi}(t,\mathbf{x}),\hat{\pi}(t,\mathbf{y})]=i\delta^{(3)}(\mathbf{x}-\mathbf{y}).$$

The remaining equal-time commutators vanish:

$$[\hat{\phi}(t,\mathbf{x}),\hat{\phi}(t,\mathbf{y})]=0,\qquad [\hat{\pi}(t,\mathbf{x}),\hat{\pi}(t,\mathbf{y})]=0.$$

These relations are the field version of

$$[\hat{q},\hat{p}]=i.$$

Instead of one coordinate and one momentum, there is one canonical pair at every point of space.

Definition

The *equal-time canonical commutation relations* are the basic operator relations between $\hat{\phi}$ and $\hat{\pi}$ at a fixed time. For bosonic fields they are commutators; for fermionic fields they will be replaced by anticommutators.

2.4 Finite Box and Continuum Normalization

Chapter 1 used a finite box to keep the momentum labels discrete. In a finite box, one writes sums over momenta and Kronecker deltas. In infinite volume, one uses integrals and Dirac deltas. The replacement rules are schematically

$$\frac{1}{V}\sum_{\mathbf{k}}\longrightarrow\int\frac{\mathrm{d}^3k}{(2\pi)^3},\qquad V\delta_{\mathbf{k},\mathbf{p}}\longrightarrow(2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}).$$

The continuum formulas below use the second convention. They look slightly more complicated than finite-box formulas, but they are the standard relativistic normalization used in scattering theory.

2.5 Mode Expansion of the Field Operator

The quantum field operator is expanded in positive- and negative-frequency solutions of the Klein–Gordon equation:

$$\hat{\phi}(t,\mathbf{x})=\int\frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\mathbf{k}}}}\left(\hat{a}_{\mathbf{k}}e^{-i\omega_{\mathbf{k}}t+i\mathbf{k}\cdot\mathbf{x}}+\hat{a}^{\dagger}_{\mathbf{k}}e^{i\omega_{\mathbf{k}}t-i\mathbf{k}\cdot\mathbf{x}}\right),$$

where

$$\omega_{\mathbf{k}}=\sqrt{\mathbf{k}^2+m^2}.$$

Since the field is real, $\hat{\phi}$ is Hermitian:

$$\hat{\phi}(t,\mathbf{x})^{\dagger}=\hat{\phi}(t,\mathbf{x}).$$

The appearance of both $\hat{a}_{\mathbf{k}}$ and $\hat{a}^{\dagger}_{\mathbf{k}}$ is the operator version of the reality condition on a classical real field.

The conjugate momentum operator is

$$\hat{\pi}(t, \mathbf{x}) = \partial_t \hat{\phi}(t, \mathbf{x}),$$

therefore

$$\hat{\pi}(t, \mathbf{x}) = -i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} t + i\mathbf{k}\cdot\mathbf{x}} - \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}} t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The normalization factors are chosen so that the equal-time field commutators are satisfied.

2.6 Ladder-Operator Algebra

The canonical field commutators imply

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \qquad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

Conversely, these ladder-operator relations imply the equal-time commutators of $\hat{\phi}$ and $\hat{\pi}$. This is the precise connection between the canonical field algebra and the oscillator algebra of Chapter 1.

The operator $\hat{a}_{\mathbf{k}}^\dagger$ creates a quantum in the momentum mode $\mathbf{k}$, while $\hat{a}_{\mathbf{k}}$ annihilates such a quantum. For the real scalar field there is only one kind of neutral scalar quantum.

2.7 Fock Space Revisited

The free-field vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for every } \mathbf{k}.$$

A one-particle momentum eigenstate is

$$|\mathbf{k}\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

A two-particle state is

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}_{\mathbf{k}_1}^\dagger \hat{a}_{\mathbf{k}_2}^\dagger|0\rangle,$$

up to normalization. Since the creation operators commute, the scalar quanta are bosons.

A realistic one-particle state is usually a wave packet rather than a single momentum eigenstate:

$$|f\rangle = \int \frac{d^3k}{(2\pi)^3} f(\mathbf{k}) \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

The function $f(\mathbf{k})$ describes the momentum profile of the state. This is the first place where the particle idea becomes more refined than the simple phrase "one quantum of one mode."

2.8 Hamiltonian and Momentum Operators

The quantum Hamiltonian is obtained from the classical Hamiltonian by replacing ϕ and π by field operators:

$$\hat{H} = \int d^3x \left(\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2 \right).$$

Substituting the mode expansion gives the formal expression

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2}(2\pi)^3 \delta^{(3)}(0) \right).$$

The first term counts excitations; the second term is the continuum notation for the sum of oscillator zero-point energies.

The spatial momentum operator is

$$\hat{\mathbf{P}} = \int \mathrm{d}^3x \, \hat{\pi}\nabla\hat{\phi}.$$

With the same ordering convention, it becomes

$$\hat{\mathbf{P}} = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, \mathbf{k} \, \hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}.$$

Thus

$$\hat{H} \, \hat{a}^\dagger_{\mathbf{k}}|0\rangle = \omega_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{k}}|0\rangle \quad \text{above the vacuum energy,}$$

and

$$\hat{\mathbf{P}} \, \hat{a}^\dagger_{\mathbf{k}}|0\rangle = \mathbf{k} \, \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

The dispersion relation is

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + m^2.$$

This is why m is interpreted as the mass of the scalar quantum.

2.9 Normal Ordering

The vacuum-energy term in $\hat{H}$ is infinite in the continuum limit. For the free theory, it is common to define the normal-ordered Hamiltonian by moving all creation operators to the left of all annihilation operators:

$$: \hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}} := \hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{k}}.$$

Then

$$: \hat{H} := \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, \omega_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}},$$

and

$$\langle 0| : \hat{H} : |0\rangle = 0.$$

Normal ordering removes the constant free vacuum energy from this Hamiltonian. It does not remove all divergences of interacting quantum field theory. It is a useful convention at the free-field level, not a substitute for renormalization.

2.10 Zero-Point Energy

In a finite box, the vacuum energy of the free scalar field is

$$E_0 = \frac{1}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

In infinite-volume notation this becomes

$$E_0 = \frac{V}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3} \, \omega_{\mathbf{k}}.$$

The integral diverges at large momentum. The divergence is a reminder that a field has infinitely many short-distance degrees of freedom.

For many non-gravitational scattering problems, only energy differences above the free vacuum matter, and normal ordering is sufficient. In contexts where vacuum energy gravitates, or where the vacuum itself changes, the issue becomes more subtle. The present course will return to this point when discussing renormalization and vacuum structure.

2.11 Fields as Operator-Valued Distributions

The notation $\hat{\phi}(x)$ suggests an operator at a point. This notation is useful, but local quantum fields are more accurately understood as operator-valued distributions. They become well-defined only after smearing with test functions.

At fixed time, a smeared field is

$$\hat{\phi}(f) = \int \mathrm{d}^3x\, f(\mathbf{x})\hat{\phi}(t, \mathbf{x}).$$

Products such as $\hat{\phi}(x)^2$ at the same spacetime point are generally singular and require a prescription. This is one of the structural reasons why renormalization appears in interacting field theory.

This point also clarifies the role of the mode expansion. The formula for $\hat{\phi}(t, \mathbf{x})$ is a compact way to encode how the field acts after it has been smeared. It should not be treated as an ordinary function with operator coefficients.

2.12 Microcausality

Relativistic quantum field theory must respect the causal structure of spacetime. For a scalar field, the basic local causality condition is

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \qquad \text{if} \qquad (x - y)^2 < 0.$$

This condition is called *microcausality*. It says that field observables associated with spacelike separated regions commute.

For the free scalar field one can write

$$[\hat{\phi}(x), \hat{\phi}(y)] = i\Delta(x - y),$$

where Δ is the Pauli–Jordan commutator function. This function vanishes outside the light cone. The equal-time relation

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0$$

is therefore only the simplest special case of the full spacetime causality condition.

Summary

Canonical quantization begins with $\hat{\phi}$ and $\hat{\pi}$, not directly with particles. The equal-time commutation relations determine the ladder operator algebra in the mode expansion. The free-field vacuum and Fock space are then recovered from that algebra, matching the oscillator picture of Chapter 1. The new lessons are the precise continuum normalization, the Hamiltonian and momentum operators, normal ordering, the distributional nature of fields, and microcausality.

2.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the classical Lagrangian and Hamiltonian of the free real scalar field;
- derive the conjugate momentum $\pi = \dot{\phi}$;
- state the equal-time canonical commutation relations;

- write the mode expansions of $\hat{\phi}$ and $\hat{\pi}$;
- derive the ladder-operator commutators from the canonical field algebra;
- construct the vacuum, one-particle states, wave packets, and many-particle states;
- express $\hat{H}$ and $\hat{\mathbf{P}}$ in terms of ladder operators;
- explain what normal ordering removes and what it does not remove;
- explain why fields are operator-valued distributions;
- state the microcausality condition for a scalar field.

Chapter 3

The Complex Scalar Field and Antiparticles

Remark

The real scalar field has one neutral quantum for each momentum mode. The complex scalar field is the next step: the field is no longer Hermitian, and its quantization produces two independent kinds of quanta. They are naturally interpreted as particles and antiparticles.

3.1 From a Real Field to a Complex Field

Chapter 2 quantized a real scalar field. The reality condition $\hat{\phi}^\dagger = \hat{\phi}$ forced the same field operator to contain both annihilation and creation operators. A complex scalar field is different. The field ϕ and its Hermitian conjugate $\phi^\dagger$ are independent classical variables before quantization.

One may think of a complex scalar field as two real scalar fields packaged into one object:

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \qquad \phi^\dagger = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2).$$

The important new feature is that the theory has a global phase symmetry. The field may be rotated by a constant phase,

$$\phi \longrightarrow e^{-i\alpha}\phi, \qquad \phi^\dagger \longrightarrow e^{i\alpha}\phi^\dagger.$$

By Noether's theorem this symmetry gives a conserved charge. In the quantum theory, that charge will distinguish particles from antiparticles.

3.2 Classical Lagrangian and Canonical Momenta

The free complex scalar field is described by

$$\mathcal{L} = \partial_\mu\phi^\dagger\partial^\mu\phi - m^2\phi^\dagger\phi.$$

In terms of time and space derivatives,

$$\mathcal{L} = \dot{\phi}^\dagger\dot{\phi} - \nabla\phi^\dagger\cdot\nabla\phi - m^2\phi^\dagger\phi.$$

The conjugate momenta are

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}^\dagger, \qquad \pi^\dagger = \frac{\partial\mathcal{L}}{\partial\dot{\phi}^\dagger} = \dot{\phi}.$$

Thus π is conjugate to ϕ, while $\pi^\dagger$ is conjugate to $\phi^\dagger$.

The equations of motion are

$$(\Box + m^2)\phi = 0, \qquad (\Box + m^2)\phi^\dagger = 0.$$

Both components satisfy the Klein–Gordon equation, but the quantum interpretation will be richer than in the real case.

3.3 Canonical Commutation Relations

Canonical quantization promotes $\phi, \phi^\dagger, \pi, \pi^\dagger$ to operators. The nonzero equal-time commutators are

$$[\hat{\phi}(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}),$$

and

$$[\hat{\phi}^\dagger(t,\mathbf{x}), \hat{\pi}^\dagger(t,\mathbf{y})] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}).$$

All other equal-time commutators vanish. In particular,

$$[\hat{\phi}(t,\mathbf{x}), \hat{\pi}^\dagger(t,\mathbf{y})] = 0, \qquad [\hat{\phi}^\dagger(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})] = 0.$$

This is the canonical field algebra for a charged bosonic scalar field.

3.4 Mode Expansion

The complex field is not Hermitian, so its mode expansion cannot be the same as the real scalar expansion. The standard continuum expansion is

$$\hat{\phi}(t,\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{k}}}\left(\hat{a}_\mathbf{k}e^{-i\omega_\mathbf{k}t+i\mathbf{k}\cdot\mathbf{x}} + \hat{b}^\dagger_\mathbf{k}e^{i\omega_\mathbf{k}t-i\mathbf{k}\cdot\mathbf{x}}\right),$$

where

$$\omega_\mathbf{k} = \sqrt{\mathbf{k}^2+m^2}.$$

The Hermitian conjugate field is

$$\hat{\phi}^\dagger(t,\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{k}}}\left(\hat{a}^\dagger_\mathbf{k}e^{i\omega_\mathbf{k}t-i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_\mathbf{k}e^{-i\omega_\mathbf{k}t+i\mathbf{k}\cdot\mathbf{x}}\right).$$

There are now two independent sets of ladder operators: the a-operators and the b-operators. This is the algebraic origin of antiparticles in the complex scalar theory.

3.5 Two Independent Oscillator Families

The nonzero ladder-operator commutators are

$$[\hat{a}_\mathbf{k}, \hat{a}^\dagger_\mathbf{p}] = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}),$$

and

$$[\hat{b}_\mathbf{k}, \hat{b}^\dagger_\mathbf{p}] = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}).$$

The two families commute with each other:

$$[\hat{a}_\mathbf{k}, \hat{b}_\mathbf{p}] = 0, \qquad [\hat{a}_\mathbf{k}, \hat{b}^\dagger_\mathbf{p}] = 0,$$

with similar relations for their Hermitian conjugates.

The vacuum is defined by

$$\hat{a}_\mathbf{k}|0\rangle = 0, \qquad \hat{b}_\mathbf{k}|0\rangle = 0 \qquad \text{for every } \mathbf{k}.$$

There are two basic one-particle states:

$$|\mathbf{k};a\rangle = \hat{a}^\dagger_\mathbf{k}|0\rangle, \qquad |\mathbf{k};b\rangle = \hat{b}^\dagger_\mathbf{k}|0\rangle.$$

They have the same mass and the same energy-momentum relation, but opposite charges.

3.6 Hamiltonian and Momentum

After normal ordering, the Hamiltonian of the free complex scalar field is

$$:\hat{H} := \int \frac{d^3k}{(2\pi)^3}\,\omega_{\mathbf{k}}\left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} + \hat{b}^\dagger_{\mathbf{k}}\hat{b}_{\mathbf{k}}\right).$$

The momentum operator is

$$:\hat{\mathbf{P}} := \int \frac{d^3k}{(2\pi)^3}\,\mathbf{k}\left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} + \hat{b}^\dagger_{\mathbf{k}}\hat{b}_{\mathbf{k}}\right).$$

Both kinds of quanta carry positive energy. The difference between them is not energy or mass, but charge.

3.7 The Charge Operator

The global phase symmetry gives a conserved current. With the convention $\phi \to e^{-i\alpha}\phi$, the corresponding quantum charge may be written, after normal ordering, as

$$\hat{Q} = \int \frac{d^3k}{(2\pi)^3}\left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} - \hat{b}^\dagger_{\mathbf{k}}\hat{b}_{\mathbf{k}}\right).$$

This convention gives

$$\hat{Q}\,\hat{a}^\dagger_{\mathbf{k}}|0\rangle = +\hat{a}^\dagger_{\mathbf{k}}|0\rangle, \qquad \hat{Q}\,\hat{b}^\dagger_{\mathbf{k}}|0\rangle = -\hat{b}^\dagger_{\mathbf{k}}|0\rangle.$$

Thus the a-quanta have charge $+1$, while the b-quanta have charge -1. They are particle and antiparticle excitations of the same complex field.

The charge operator also generates the phase transformation of the field:

$$[\hat{Q}, \hat{\phi}(x)] = -\hat{\phi}(x), \qquad [\hat{Q}, \hat{\phi}^\dagger(x)] = +\hat{\phi}^\dagger(x).$$

This is the operator statement that $\hat{\phi}$ carries charge -1 as an operator, even though $\hat{a}^\dagger$ creates a charge $+1$ state in the chosen convention.

3.8 Positive and Negative Frequency Revisited

In the real scalar field, the negative-frequency part was simply the Hermitian conjugate of the positive-frequency part of the same field. In the complex field, the negative-frequency part of $\hat{\phi}$ contains $\hat{b}^\dagger$, not $\hat{a}^\dagger$. This is the crucial difference.

The field $\hat{\phi}$ can annihilate an a-particle or create a b-antiparticle. The conjugate field $\hat{\phi}^\dagger$ can create an a-particle or annihilate a b-antiparticle. Symbolically,

$$\hat{\phi} \sim \hat{a} + \hat{b}^\dagger, \qquad \hat{\phi}^\dagger \sim \hat{a}^\dagger + \hat{b}.$$

This is one of the cleanest ways to see why relativistic quantum fields contain antiparticles.

3.9 Why Antiparticles Appear Naturally

The appearance of antiparticles is not an optional addition. It follows from combining three requirements:

- relativistic positive-energy modes;
- canonical quantization;

- a complex field with a conserved charge.

The field must contain both positive- and negative-frequency solutions in order to be local and to satisfy the canonical commutation relations. For a charged field, those two frequency sectors are associated with two independent oscillator families. The quanta created by the second family have the opposite charge and are interpreted as antiparticles.

3.10 Complex Scalar Propagator

The natural two-point function of the complex scalar field is

$$D_F(x-y) = \langle 0|T\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle.$$

It has the same momentum-space denominator as the real scalar propagator:

$$D_F(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

The difference is in the charge flow. The propagator connects ϕ with $\phi^\dagger$, because charge must be conserved. Correlators such as $\langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ vanish in the free theory when the vacuum has zero charge.

3.11 Physical Interpretation of the Fock Space

The Fock space of the complex scalar field contains particles, antiparticles, and arbitrary multiparticle states. Examples are

$$\hat{a}^\dagger_{\mathbf{k}}|0\rangle, \qquad \hat{b}^\dagger_{\mathbf{p}}|0\rangle, \qquad \hat{a}^\dagger_{\mathbf{k}}\hat{b}^\dagger_{\mathbf{p}}|0\rangle.$$

The last state contains one particle and one antiparticle. Its total charge is zero, but it is not the vacuum. Charge conservation restricts which transitions are possible once interactions are introduced.

Summary

A complex scalar field is equivalent to two real scalar degrees of freedom, but its global phase symmetry gives a conserved charge. Quantization produces two independent oscillator families. One creates particles and the other creates antiparticles, with equal mass and opposite charge. The field $\hat{\phi}$ contains an annihilation operator for particles and a creation operator for antiparticles, while $\hat{\phi}^\dagger$ does the reverse.

3.12 Chapter Checklist

After reading this chapter, one should be able to:

- explain why ϕ and $\phi^\dagger$ are independent field variables before quantization;
- write the canonical momenta of the complex scalar field;
- state the equal-time commutation relations;
- write the mode expansions of $\hat{\phi}$ and $\hat{\phi}^\dagger$;
- explain why two independent ladder-operator families appear;

- construct particle and antiparticle states;
- write the charge operator and interpret its eigenvalues;
- explain why antiparticles are required in a relativistic charged field;
- identify the charge structure of the complex scalar propagator.

Chapter 4

Quantization of the Dirac Field

Remark

Scalar fields describe spin-zero quanta and are quantized with commutators. The Dirac field describes spin-1/2 quanta. Its consistent quantization requires anticommutators, and its mode expansion naturally contains particle and antiparticle operators with spin labels.

4.1 Why Fermionic Quantization Is Different

Chapters 2 and 3 quantized bosonic scalar fields. The ladder operators obeyed commutation relations, and many-particle states were symmetric under exchange. The Dirac field is different because its quanta are fermions. Fermionic states must be antisymmetric under exchange, and no two identical fermions can occupy the same one-particle state.

This difference is implemented algebraically by replacing commutators with anticommutators. For operators $\hat{A}$ and $\hat{B}$, the anticommutator is

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

The quantized Dirac field will be built from creation and annihilation operators whose anticommutation relations enforce Fermi statistics.

There is also a second important difference. The Dirac equation is first order in time derivatives, unlike the Klein–Gordon equation. This changes the canonical structure and makes the Dirac field look less like a collection of ordinary oscillator coordinates. Nevertheless, the final Fock-space structure is parallel to the scalar case: there are annihilation operators, creation operators, a vacuum, particles, antiparticles, and number operators.

4.2 Classical Dirac Field

The Dirac field is a spinor field $\psi(x)$. Its Dirac adjoint is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

The free Dirac Lagrangian density is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

The equation of motion is the Dirac equation,

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.$$

The adjoint field satisfies

$$\bar{\psi}(x)(i\overleftarrow{\partial}_\mu \gamma^\mu + m) = 0,$$

where the arrow indicates that the derivative acts to the left.

The matrices γ^μ obey the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}.$$

This algebra is what makes the Dirac equation compatible with the relativistic mass-shell relation $p^2 = m^2$.

4.3 Canonical Structure and Anticommutation Relations

Because the Lagrangian is first order in $\dot{\psi}$, the momentum conjugate to ψ is proportional to $\psi^\dagger$:

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger.$$

The quantum theory is defined by equal-time anticommutation relations,

$$\{\hat{\psi}_\alpha(t,\mathbf{x}), \hat{\psi}^\dagger_\beta(t,\mathbf{y})\} = \delta_{\alpha\beta}\delta^{(3)}(\mathbf{x}-\mathbf{y}),$$

with

$$\{\hat{\psi}_\alpha(t,\mathbf{x}), \hat{\psi}_\beta(t,\mathbf{y})\} = 0, \qquad \{\hat{\psi}^\dagger_\alpha(t,\mathbf{x}), \hat{\psi}^\dagger_\beta(t,\mathbf{y})\} = 0.$$

The spinor indices α, β label the components of the Dirac spinor.

Remark

The replacement of commutators by anticommutators is not a cosmetic change. It is what produces fermionic Fock space, positive-energy particle interpretation, and the Pauli exclusion principle.

4.4 Mode Expansion of the Dirac Field

The free Dirac equation has positive-energy spinor solutions $u_s(\mathbf{p})$ and negative-frequency spinor solutions $v_s(\mathbf{p})$, where s labels spin. A standard expansion is

$$\hat{\psi}(x) = \sum_s \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_\mathbf{p}}}\left(\hat{b}_s(\mathbf{p})u_s(\mathbf{p})e^{-iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}} + \hat{d}^\dagger_s(\mathbf{p})v_s(\mathbf{p})e^{iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}}\right),$$

where

$$E_\mathbf{p} = \sqrt{\mathbf{p}^2 + m^2}.$$

The adjoint field is

$$\hat{\bar{\psi}}(x) = \sum_s \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_\mathbf{p}}}\left(\hat{b}^\dagger_s(\mathbf{p})\bar{u}_s(\mathbf{p})e^{iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s(\mathbf{p})\bar{v}_s(\mathbf{p})e^{-iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}}\right).$$

The structure is analogous to the complex scalar field:

$$\hat{\psi} \sim \hat{b} + \hat{d}^\dagger, \qquad \hat{\bar{\psi}} \sim \hat{b}^\dagger + \hat{d}.$$

The operators $\hat{b}^\dagger$ create particles, while $\hat{d}^\dagger$ create antiparticles.

4.5 Spinor Creation and Annihilation Operators

The nonzero anticommutators are

$$\{\hat{b}_s(\mathbf{p}), \hat{b}^\dagger_r(\mathbf{q})\} = (2\pi)^3\delta_{sr}\delta^{(3)}(\mathbf{p}-\mathbf{q}),$$

and

$$\{\hat{d}_s(\mathbf{p}), \hat{d}^\dagger_r(\mathbf{q})\} = (2\pi)^3\delta_{sr}\delta^{(3)}(\mathbf{p}-\mathbf{q}).$$

All other anticommutators vanish. In particular,

$$\{\hat{b}_s(\mathbf{p}), \hat{d}_r(\mathbf{q})\} = 0, \qquad \{\hat{b}_s(\mathbf{p}), \hat{d}^\dagger_r(\mathbf{q})\} = 0.$$

The label s is the spin label. For a massive Dirac particle there are two spin states for the particle and two spin states for the antiparticle.

Because the creation operators anticommute, applying the same creation operator twice gives zero:

$$\left(\hat{b}_s^\dagger(\mathbf{p})\right)^2 = 0, \qquad \left(\hat{d}_s^\dagger(\mathbf{p})\right)^2 = 0.$$

This is the algebraic form of the Pauli exclusion principle.

4.6 Particle and Antiparticle States

The vacuum is defined by

$$\hat{b}_s(\mathbf{p})|0\rangle = 0, \qquad \hat{d}_s(\mathbf{p})|0\rangle = 0 \qquad \text{for all } \mathbf{p}, s.$$

A one-particle state is

$$|\mathbf{p}, s; b\rangle = \hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

A one-antiparticle state is

$$|\mathbf{p}, s; d\rangle = \hat{d}_s^\dagger(\mathbf{p})|0\rangle.$$

For the electron field, the $b^\dagger$-states are electron states and the $d^\dagger$-states are positron states. More generally, a Dirac field creates and annihilates a charged fermion and its antiparticle.

A two-fermion state changes sign when the order of the creation operators is exchanged:

$$\hat{b}_s^\dagger(\mathbf{p})\hat{b}_r^\dagger(\mathbf{q})|0\rangle = -\hat{b}_r^\dagger(\mathbf{q})\hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

This antisymmetry is the defining feature of fermionic Fock space.

4.7 Hamiltonian, Momentum, and Charge

After normal ordering, the free Dirac Hamiltonian is

$$:\hat{H}: = \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3} E_{\mathbf{p}} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

The momentum operator is

$$:\hat{\mathbf{P}}: = \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \mathbf{p} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

Both particles and antiparticles carry positive energy.

For a Dirac field with unit charge convention, the normal-ordered charge has the form

$$\hat{Q} = \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) - \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

For the electron field one often multiplies this operator by $-e$, so that particles created by $\hat{b}^\dagger$ have charge $-e$, and antiparticles created by $\hat{d}^\dagger$ have charge $+e$.

4.8 Why Anticommutators Give the Correct Energy Spectrum

The negative-frequency part of the Dirac field is not interpreted as a tower of negative-energy particles. After quantization, it is reinterpreted as the creation of antiparticles with positive energy. The anticommutation relations are essential for this reinterpretation to give a Hamiltonian bounded below.

Schematically, the Dirac Hamiltonian before normal ordering contains terms of the form

$$\hat{b}^\dagger \hat{b} - \hat{d}\hat{d}^\dagger.$$

Using anticommutators,

$$\hat{d}\hat{d}^\dagger = 1 - \hat{d}^\dagger \hat{d},$$

so, up to an infinite constant that is removed by normal ordering, the antiparticle contribution becomes positive:

$$\hat{d}^\dagger \hat{d}.$$

This is the algebraic reason the free Dirac Hamiltonian counts both particles and antiparticles with positive energy.

4.9 Spinor Sums and Completeness

The spinor wave functions are normalized so that they satisfy completeness relations. A common convention is

$$\sum_s u_s(\mathbf{p})\bar{u}_s(\mathbf{p}) = \gamma^\mu p_\mu + m,$$

and

$$\sum_s v_s(\mathbf{p})\bar{v}_s(\mathbf{p}) = \gamma^\mu p_\mu - m,$$

where $p^0 = E_{\mathbf{p}}$. These identities are used constantly when computing fermion propagators and scattering amplitudes.

The spinor sums play for the Dirac field a role analogous to polarization sums for vector fields. They allow spin labels on internal lines to be summed in a Lorentz-covariant way.

4.10 The Dirac Propagator

The Feynman propagator of the Dirac field is

$$S_F(x-y) = \langle 0|T\hat{\psi}(x)\hat{\bar{\psi}}(y)|0\rangle.$$

In momentum space it is

$$S_F(x-y) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)}.$$

Equivalently, the propagator is the Green function of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m)S_F(x-y) = i\delta^{(4)}(x-y).$$

Compared with the scalar propagator, the denominator is the same mass-shell factor, but the numerator carries spinor structure.

4.11 Chirality, Helicity, and the Massless Limit

The Dirac field contains spin degrees of freedom. For massive fermions, the spin label may be chosen in several equivalent ways. For massless fermions, helicity and chirality become especially important.

The chirality matrix γ^5 defines projection operators

$$P_L = \frac{1}{2}(1-\gamma^5), \qquad P_R = \frac{1}{2}(1+\gamma^5).$$

They split a Dirac field into left- and right-chiral components:

$$\psi_L = P_L\psi, \qquad \psi_R = P_R\psi.$$

For a massless Dirac field, the left- and right-chiral parts decouple. This will be essential later in the electroweak theory, where left- and right-handed fermions transform differently under gauge symmetries.

4.12 Majorana Fields: Conceptual Overview

A Dirac field describes a charged fermion and a distinct antiparticle. A Majorana field is different: it is a fermionic field that is equal to its charge conjugate in an appropriate sense. Informally, a Majorana fermion is its own antiparticle.

The Majorana idea is possible for neutral fermions, not for fields carrying an ordinary conserved electric charge. The full construction requires charge conjugation matrices and reality conditions on spinors. For now, the important contrast is simple:

$$\text{Dirac field: particle and antiparticle are distinct,}$$

while

$$\text{Majorana field: particle and antiparticle are identified.}$$

Majorana fields will become relevant when discussing neutrino masses and extensions of the Standard Model.

Summary

The Dirac field is the first fermionic quantum field in the course. Its quantization uses anticommutators rather than commutators. The mode expansion contains particle annihilation operators and antiparticle creation operators, now with spin labels. Anticommutation gives fermionic Fock space, the Pauli exclusion principle, and a Hamiltonian with positive particle and antiparticle energies after normal ordering.

4.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the free Dirac Lagrangian and Dirac equation;
- explain why fermionic fields are quantized with anticommutators;
- write the mode expansion of $\hat{\psi}$ and $\hat{\bar{\psi}}$;
- state the anticommutation relations of $\hat{b}, \hat{b}^\dagger$ and $\hat{d}, \hat{d}^\dagger$;
- construct fermion and antifermion states;
- explain how the Hamiltonian counts positive-energy particles and antiparticles;
- write the charge operator and interpret the sign convention;
- state the standard spinor completeness relations;
- write the Dirac propagator;
- distinguish Dirac and Majorana fermions at the conceptual level.

Chapter 5

Quantization of the Electromagnetic Field

Remark

The scalar and Dirac fields taught us how particles and antiparticles arise from mode expansions. The electromagnetic field adds a new feature: gauge redundancy. A photon is not an excitation of four independent components of A_μ, but of the physical transverse degrees of freedom of a gauge field.

5.1 Why Gauge Fields Are Harder to Quantize

The real scalar, complex scalar, and Dirac fields all had independent field components that could be quantized after imposing the appropriate commutation or anticommutation relations. The electromagnetic field is different. It is described by a vector potential A_μ, but not all components of A_μ represent physical degrees of freedom.

The field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The free electromagnetic Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

It is invariant under the gauge transformation

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda.$$

The potentials A_μ and $A_\mu + \partial_\mu \Lambda$ describe the same physical electromagnetic field. Therefore quantizing every component of A_μ as if it were an independent scalar field would overcount states.

5.2 Physical and Redundant Degrees of Freedom

A massless vector field in four spacetime dimensions has two physical polarizations. This is already suggested classically by electromagnetic waves: the electric and magnetic fields of a plane wave are transverse to the direction of propagation.

The four components of A_μ contain redundant information. Gauge freedom removes one part, and the equations of motion impose a constraint. The result is that only two transverse photon polarizations are physical.

This distinction is central:

$$\text{four components of } A_\mu \neq \text{four physical photon polarizations.}$$

The task of quantization is to keep Lorentz covariance and locality under control while ensuring that only the physical states contribute to observable processes.

5.3 Canonical Obstruction

The canonical momentum conjugate to A_μ is

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}.$$

For the time component one finds

$$\Pi^0 = 0.$$

This means that A_0 has no independent time derivative in the Lagrangian. It is not an ordinary dynamical coordinate. Instead, it acts as a Lagrange multiplier imposing Gauss's law.

This is the first signal that gauge fields require constraints or gauge fixing. One cannot simply impose four independent canonical commutation relations for all components of A_μ.

5.4 Coulomb Gauge Quantization

A physically transparent gauge choice is Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0.$$

For the free electromagnetic field, one may also set the nondynamical scalar potential to zero after solving the constraint. The remaining field is the transverse vector potential $\mathbf{A}_T$.

The transverse field may be expanded as

$$\hat{\mathbf{A}}(t, \mathbf{x}) = \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{\sqrt{2|\mathbf{k}|}} \left(\epsilon_\lambda(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}) e^{-i|\mathbf{k}|t + i\mathbf{k}\cdot\mathbf{x}} + \epsilon_\lambda^*(\mathbf{k}) \hat{a}_\lambda^\dagger(\mathbf{k}) e^{i|\mathbf{k}|t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The polarization vectors are transverse:

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0.$$

The index $\lambda = 1, 2$ labels the two physical polarizations.

5.5 Photon Creation and Annihilation Operators

The photon ladder operators satisfy

$$[\hat{a}_\lambda(\mathbf{k}), \hat{a}_\sigma^\dagger(\mathbf{p})] = (2\pi)^3 \delta_{\lambda\sigma} \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with all other commutators vanishing. The vacuum is defined by

$$\hat{a}_\lambda(\mathbf{k})|0\rangle = 0 \qquad \text{for all } \mathbf{k}, \lambda.$$

A one-photon state is

$$|\mathbf{k}, \lambda\rangle = \hat{a}_\lambda^\dagger(\mathbf{k})|0\rangle.$$

Its energy is

$$\omega_{\mathbf{k}} = |\mathbf{k}|,$$

which is the massless dispersion relation.

The normal-ordered Hamiltonian is

$$: \hat{H} := \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} |\mathbf{k}| \, \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

The momentum operator is

$$: \hat{\mathbf{P}} := \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathbf{k} \, \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

5.6 Polarization Vectors

For each nonzero momentum $\mathbf{k}$, one chooses two transverse polarization vectors. They obey

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0, \qquad \epsilon^*_\lambda(\mathbf{k}) \cdot \epsilon_\sigma(\mathbf{k}) = \delta_{\lambda\sigma}.$$

Linear polarizations are useful for many classical comparisons. Circular polarizations are useful for describing helicity. They are built from two linear polarizations by

$$\epsilon_\pm = \frac{1}{\sqrt{2}} (\epsilon_1 \pm i\epsilon_2) .$$

The labels $+$ and $-$ correspond to the two helicity states of the photon.

5.7 Photon States and Helicity

A massive spin-one particle would have three spin polarizations. A photon is massless and gauge invariant, so only two physical helicities remain. A one-photon state may be labelled as

$$|\mathbf{k}, +\rangle, \qquad |\mathbf{k}, -\rangle.$$

These are helicity eigenstates. Helicity is the projection of angular momentum onto the direction of motion. For a massless particle, helicity is Lorentz invariant under proper orthochronous Lorentz transformations.

The absence of a longitudinal physical photon is not an accident. The longitudinal component of $\mathbf{A}$ is gauge redundancy, not a separate particle.

5.8 Covariant Quantization

Coulomb gauge makes the physical degrees of freedom visible, but it hides manifest Lorentz covariance. For perturbative QFT it is often more convenient to use a covariant gauge-fixing term,

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The gauge-fixed Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The parameter ξ labels a family of gauges. The choice $\xi = 1$ is called Feynman gauge.

Covariant quantization introduces unphysical components into intermediate calculations. These components must not appear in physical observables. In QED, current conservation ensures that unphysical polarizations cancel from scattering amplitudes. In non-Abelian gauge theory, the corresponding statement requires ghosts and BRST symmetry, which will be developed later.

5.9 Gauge Fixing and the Photon Propagator

In covariant gauges, the momentum-space photon propagator is

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2 + i\epsilon} \right) .$$

In Feynman gauge, this simplifies to

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The pole at $k^2 = 0$ reflects the masslessness of the photon. The numerator carries the vector structure of the field.

The gauge-dependent part proportional to $k_\mu k_\nu$ does not change physical QED amplitudes coupled to conserved currents. This is the practical reason covariant gauges are useful: they simplify calculations without changing observable predictions.

5.10 Unphysical Polarizations

Covariant quantization treats A_μ as a four-component field during intermediate steps. This introduces time-like and longitudinal polarizations that are not physical photon states. The physical Hilbert space is obtained only after imposing the appropriate condition that removes the unphysical sector.

In elementary QED calculations, this removal is often hidden inside Ward identities and current conservation. If an external photon polarization vector is shifted by a multiple of its momentum,

$$\epsilon_\mu(k) \longrightarrow \epsilon_\mu(k) + c\, k_\mu,$$

physical amplitudes are unchanged. This is the amplitude-level expression of gauge redundancy.

5.11 Gauge-Invariant Observables

The potential A_μ is not gauge invariant. The field strength $F_{\mu\nu}$ is gauge invariant, and so are the electric and magnetic fields. Observables must ultimately be built from gauge-invariant quantities or from gauge-invariant combinations of charged fields and gauge fields.

Examples include

$$F_{\mu\nu}(x), \qquad F_{\mu\nu}(x)F^{\mu\nu}(x),$$

and, in more advanced settings, Wilson line and Wilson loop operators. The important lesson at this stage is that the photon is described by a gauge potential, but physical information cannot depend on the redundant gauge choice.

5.12 What Is a Photon?

In free QED, a photon is a one-quantum excitation of a transverse mode of the electromagnetic field:

$$|\mathbf{k}, \lambda\rangle = \hat{a}^\dagger_\lambda(\mathbf{k})|0\rangle, \qquad \lambda = 1, 2.$$

Equivalently, one may label the two states by helicity $\lambda = \pm$. The photon is massless, has energy $|\mathbf{k}|$, and carries no electric charge.

This definition is sharp in the free theory and in scattering theory with asymptotic photon states. In interacting QED, soft photons and infrared effects make the exact particle interpretation more subtle. Those subtleties will be revisited when discussing scattering and loop corrections.

Summary

The electromagnetic field is the first gauge field in the course. Gauge redundancy means that not all components of A_μ are physical. In Coulomb gauge the two transverse photon polarizations are explicit. In covariant gauges one keeps Lorentz covariance at the cost of introducing unphysical components, which cancel from physical observables. A photon is a transverse, massless quantum of the electromagnetic field.

5.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain why gauge fields are harder to quantize than scalar fields;
- state the gauge redundancy of A_μ;
- explain why the photon has two physical polarizations;
- write the Coulomb-gauge mode expansion of the transverse vector potential;
- construct one-photon states;
- distinguish linear polarization from helicity;
- write the covariant gauge-fixing term;
- write the photon propagator in a covariant gauge;
- explain why unphysical polarizations do not appear as external photon states;
- identify gauge-invariant electromagnetic observables.

Chapter 6

Propagators as Quantum Correlation Functions

Remark

The previous chapters constructed free quantum fields and their particle states. This chapter introduces the next central object: the propagator. A propagator is not merely a line in a Feynman diagram. It is a vacuum correlation function of quantum fields, and it encodes both the equation of motion and the particle content of the theory.

6.1 From Classical Green Functions to Quantum Correlators

In classical field theory, a Green function solves an inhomogeneous differential equation. For the Klein–Gordon operator, one writes

$$(\Box_x + m^2)G(x-y) = \delta^{(4)}(x-y).$$

Different boundary conditions give different Green functions: retarded, advanced, or other choices.

In quantum field theory, a closely related object appears, but with a different interpretation. The basic two-point function is a vacuum expectation value of a product of field operators. For a real scalar field, the simplest one is

$$\langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

This is not the value of a classical field at two points. It is a quantum correlation between two field operators in the vacuum.

6.2 Wightman Functions

The positive-frequency Wightman function of the real scalar field is

$$\Delta^+(x-y) = \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

Using the mode expansion, one finds

$$\Delta^+(x-y) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{2\omega_{\mathbf{k}}}e^{-i\omega_{\mathbf{k}}(x^0-y^0)+i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}.$$

The reversed ordering gives

$$\Delta^-(x-y) = \langle 0|\hat{\phi}(y)\hat{\phi}(x)|0\rangle.$$

These functions measure vacuum correlations. Even in the vacuum, field operators at different spacetime points are correlated.

The commutator function is the difference

$$[\hat{\phi}(x), \hat{\phi}(y)] = \Delta^+(x-y) - \Delta^-(x-y) = i\Delta(x-y).$$

Microcausality says that this commutator vanishes for spacelike separation.

6.3 Time-Ordered Products

Scattering theory is built from time-ordered products. For two bosonic field operators, the time-ordered product is defined by

$$T\hat{\phi}(x)\hat{\phi}(y) = \begin{cases} \hat{\phi}(x)\hat{\phi}(y), & x^0 > y^0, \\ \hat{\phi}(y)\hat{\phi}(x), & y^0 > x^0. \end{cases}$$

Equivalently,

$$T\hat{\phi}(x)\hat{\phi}(y) = \theta(x^0 - y^0)\hat{\phi}(x)\hat{\phi}(y) + \theta(y^0 - x^0)\hat{\phi}(y)\hat{\phi}(x).$$

For fermionic fields, time ordering includes a minus sign when two fermion operators are exchanged. This sign is one of the first places where fermionic statistics enters perturbation theory.

6.4 The Feynman Propagator

The scalar Feynman propagator is the vacuum expectation value of the time-ordered two-point function:

$$D_F(x - y) = \langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

Using the Wightman functions, this may be written as

$$D_F(x - y) = \theta(x^0 - y^0)\Delta^+(x - y) + \theta(y^0 - x^0)\Delta^+(y - x).$$

It has the four-dimensional momentum-space representation

$$D_F(x - y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

The small $i\epsilon$ prescription specifies how the poles are passed in the complex k^0-plane. This prescription is what makes the propagator the vacuum time-ordered correlation function rather than some other Green function.

6.5 The Propagator as a Green Function

The Feynman propagator satisfies

$$(\Box_x + m^2)D_F(x - y) = -i\delta^{(4)}(x - y),$$

with the sign convention following from the definition above. Thus the Feynman propagator is also a Green function of the Klein–Gordon operator, but with a specific quantum boundary condition.

This dual nature is important:

$$\text{propagator} = \text{Green function} = \text{vacuum two-point correlation function.}$$

The first viewpoint emphasizes equations of motion. The second emphasizes quantum states and operator expectation values.

6.6 Retarded and Advanced Propagators

The retarded propagator is built from the commutator:

$$D_R(x - y) = i\theta(x^0 - y^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 < y^0$, so it describes causal response to a disturbance in the past. The advanced propagator is

$$D_A(x-y) = -i\theta(y^0 - x^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 > y^0$.

Retarded and advanced propagators are especially natural in classical response problems and nonequilibrium physics. The Feynman propagator is the object that appears in ordinary perturbative scattering theory.

6.7 The $i\epsilon$ Prescription

The expression

$$\frac{i}{k^2 - m^2 + i\epsilon}$$

contains more information than the algebraic denominator. The poles are located near

$$k^0 = \pm\omega_{\mathbf{k}}.$$

The $i\epsilon$ prescription tells us how to deform the integration contour around these poles. It is the momentum-space encoding of time ordering.

For positive time separation, the contour picks up the positive-energy pole. For negative time separation, it picks up the negative-energy pole in the way appropriate to a time-ordered vacuum expectation value. This is how the compact four-dimensional expression reproduces the two time orderings in position space.

6.8 Momentum-Space Propagators for Free Fields

For the real scalar field,

$$D_F(k) = \frac{i}{k^2 - m^2 + i\epsilon}.$$

For the complex scalar field,

$$\langle 0|T\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle$$

has the same momentum-space denominator. The difference is that the complex scalar propagator carries charge flow from $\phi^\dagger$ to ϕ.

For the Dirac field,

$$S_F(k) = \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}.$$

The denominator is the same mass-shell structure, while the numerator carries spinor information.

For the photon in Feynman gauge,

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

In a general covariant gauge, the numerator changes by gauge-dependent terms. Physical amplitudes do not depend on that gauge choice.

6.9 Pole Structure and Particle Interpretation

The location of poles in a propagator reveals the particle content of the free theory. The scalar propagator has poles at

$$k^2 = m^2,$$

which is the relativistic mass shell. This is the analytic trace of a particle of mass m.

For a massless photon, the pole is at

$$k^2 = 0.$$

For a free Dirac field, the denominator again has poles at $k^2 = m^2$, while the numerator encodes spinor structure.

In an interacting theory, the pole structure becomes even more informative. A shift in the pole position corresponds to a mass correction. A pole with an imaginary part corresponds to an unstable resonance. Branch cuts correspond to multiparticle thresholds. These ideas will be developed later, but the free propagator already contains the basic principle: particles appear as singular structures of correlation functions.

6.10 Propagation Versus Correlation

The word "propagator" can be misleading. It suggests that a particle literally travels from y to x. In QFT the primary object is a correlation function of fields. It measures the amplitude associated with inserting one field at y and another at x, with the vacuum in between.

For spacelike separation, the Feynman propagator need not vanish. This does not violate causality, because the commutator vanishes outside the light cone. The causal statement is not that all vacuum correlations vanish at spacelike separation, but that spacelike separated observables commute.

Thus one should distinguish

$$\text{vacuum correlation} \neq \text{signal propagation}.$$

The propagator is a correlation function used in the computation of amplitudes; it is not by itself a directly observable signal.

6.11 What a Propagator Means in Perturbation Theory

In perturbation theory, propagators connect field insertions. Wick's theorem will show that time-ordered products of free fields can be reduced to sums of contractions, and each contraction is a Feynman propagator.

This is why propagators become internal lines in Feynman diagrams. A diagram is a compact bookkeeping device for a term in the perturbative expansion. The line stands for a two-point function of the corresponding free field.

For example, a scalar internal line carries

$$\frac{i}{k^2 - m^2 + i\epsilon},$$

a fermion internal line carries

$$\frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon},$$

and a photon internal line in Feynman gauge carries

$$\frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The diagrammatic rules are therefore not arbitrary. They are built from the free-field correlation functions developed in this chapter.

6.12 Summary of the Main Propagators

The most important free propagators introduced so far are

$$
\begin{aligned}
&\text{real scalar} &:\quad& \frac{i}{k^2 - m^2 + i\epsilon}, \\
&\text{complex scalar} &:\quad& \frac{i}{k^2 - m^2 + i\epsilon} \text{ with charge flow}, \\
&\text{Dirac field} &:\quad& \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}, \\
&\text{photon, Feynman gauge} &:\quad& \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.
\end{aligned}
$$

The denominators encode the mass shell. The numerators encode spin, vector, or gauge structure.

Summary

A propagator is a time-ordered vacuum two-point function. It is also a Green function with a specific quantum boundary condition. Its poles encode particle content, while its numerator encodes spin and gauge structure. Propagators are the building blocks of perturbative QFT because Wick's theorem turns contractions of free fields into propagators.

6.13 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish classical Green functions from quantum two-point functions;
- define Wightman functions;
- define the time-ordered product;
- write the scalar Feynman propagator in position and momentum space;
- explain the role of the $i\epsilon$ prescription;
- distinguish Feynman, retarded, and advanced propagators;
- write the free scalar, Dirac, and photon propagators;
- explain how pole structure encodes particle content;
- explain why vacuum correlation is not the same as signal propagation;
- explain why propagators become internal lines in Feynman diagrams.

Chapter 7

The Interaction Picture

Remark

The previous chapter introduced propagators of free fields. Perturbation theory uses those free fields as the starting point and treats interactions as a controlled deformation. The interaction picture is the framework that makes this separation precise.

7.1 Why We Need a New Picture

In quantum mechanics one may describe time evolution in several equivalent ways. The Schroedinger picture puts time dependence in the states. The Heisenberg picture puts time dependence in the operators. The interaction picture splits the time dependence between them.

This split is useful when the Hamiltonian has the form

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}.$$

Here $\hat{H}_0$ is the exactly solvable free Hamiltonian, and $\hat{H}_{\text{int}}$ contains interactions. In QFT, $\hat{H}_0$ is built from the free scalar, spinor, or photon fields of the previous chapters. The interaction part describes processes such as scattering, decay, emission, and absorption.

The guiding idea is simple:

solve the free theory exactly, then expand in the interaction.

The interaction picture is the operator language in which this idea becomes a systematic perturbation theory.

7.2 Schroedinger, Heisenberg, and Interaction Pictures

Let $|\Psi_S(t)\rangle$ be a Schroedinger-picture state. It satisfies

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\Psi_S(t)\rangle = \hat{H}|\Psi_S(t)\rangle.$$

Operators in this picture are usually time independent unless they have explicit time dependence.

In the Heisenberg picture, the state is fixed and operators evolve according to

$$\hat{O}_H(t) = e^{i\hat{H}t}\hat{O}_S e^{-i\hat{H}t}.$$

This picture is elegant but difficult for interacting QFT because exact interacting operators are rarely known.

In the interaction picture, free time evolution is assigned to the operators:

$$\hat{O}_I(t) = e^{i\hat{H}_0 t}\hat{O}_S e^{-i\hat{H}_0 t}.$$

The states evolve only with the interaction. Thus interaction-picture fields are free fields, even though the states know about the interaction.

7.3 Free Fields as the Starting Point

The scalar interaction-picture field satisfies the free Klein–Gordon equation,

$$(\Box + m^2)\hat{\phi}_I(x) = 0.$$

Its mode expansion is the same free expansion used in Chapter 2:

$$\hat{\phi}_I(x) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-ik\cdot x} + \hat{a}^\dagger_{\mathbf{k}} e^{ik\cdot x} \right),$$

where $k^0 = \omega_{\mathbf{k}}$. Similarly, interaction-picture Dirac and photon fields are free Dirac and free photon fields.

This is why the propagators of Chapter 6 are the basic building blocks of perturbation theory. The fields appearing inside the perturbative expansion are free fields, so their time-ordered products can be evaluated using free-field correlation functions.

7.4 The Interaction Hamiltonian

The interaction Hamiltonian in the interaction picture is

$$\hat{H}_I(t) = e^{i\hat{H}_0 t} \hat{H}_{\text{int}} e^{-i\hat{H}_0 t}.$$

It is written in terms of interaction-picture fields. For example, a scalar ϕ^4 interaction is described by

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

For this case, the interaction Hamiltonian density is

$$\mathcal{H}_{\text{int}} = \frac{\lambda}{4!}\phi_I^4,$$

and

$$\hat{H}_I(t) = \int \mathrm{d}^3x \, \frac{\lambda}{4!} \hat{\phi}_I(t, \mathbf{x})^4.$$

The subscript I reminds us that the fields evolve freely. The coupling λ measures the strength of the interaction.

For QED, the interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu,$$

so the interaction Hamiltonian is commonly written as

$$\hat{H}_I(t) = e \int \mathrm{d}^3x \, \hat{\bar{\psi}}_I \gamma^\mu \hat{\psi}_I \hat{A}_{I\mu}.$$

This term is responsible for the elementary QED vertex.

7.5 Time Evolution Operator

Interaction-picture states satisfy

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\Psi_I(t)\rangle = \hat{H}_I(t)|\Psi_I(t)\rangle.$$

Define the time evolution operator $\hat{U}_I(t, t_0)$ by

$$|\Psi_I(t)\rangle = \hat{U}_I(t, t_0)|\Psi_I(t_0)\rangle.$$

Then $\hat{U}_I(t, t_0)$ obeys

$$i\frac{\partial}{\partial t}\hat{U}_I(t, t_0) = \hat{H}_I(t)\hat{U}_I(t, t_0), \qquad \hat{U}_I(t_0, t_0) = 1.$$

Formally, this is solved by a time-ordered exponential:

$$\hat{U}_I(t, t_0) = T \exp\left(-i\int_{t_0}^{t} \mathrm{d}t'\, \hat{H}_I(t')\right).$$

The time-ordering symbol is essential because $\hat{H}_I(t_1)$ and $\hat{H}_I(t_2)$ need not commute at different times.

7.6 The Dyson Series

Expanding the time-ordered exponential gives the Dyson series:

$$\hat{U}_I(t, t_0) = 1 - i\int_{t_0}^{t} \mathrm{d}t_1\, \hat{H}_I(t_1) + \frac{(-i)^2}{2!}\int_{t_0}^{t} \mathrm{d}t_1\, \mathrm{d}t_2\, T\{\hat{H}_I(t_1)\hat{H}_I(t_2)\} + \cdots.$$

Equivalently,

$$\hat{U}_I(t, t_0) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!}\int_{t_0}^{t} \mathrm{d}t_1 \cdots \mathrm{d}t_n\, T\{\hat{H}_I(t_1)\cdots\hat{H}_I(t_n)\}.$$

Each power of $\hat{H}_I$ brings one power of the coupling. In ϕ^4 theory, the n-th order term contains λ^n. In QED, the n-th order term contains e^n.

This is the algebraic origin of perturbation theory: physical quantities are computed as series in coupling constants.

7.7 Perturbation Theory and Time-Ordered Correlators

Suppose we want the interacting time-ordered two-point function of a scalar field. In the interaction picture it is expressed as

$$\langle\Omega|T\hat{\phi}_H(x)\hat{\phi}_H(y)|\Omega\rangle = \frac{\langle 0|T\left\{\hat{\phi}_I(x)\hat{\phi}_I(y)\exp\left[-i\int \mathrm{d}^4z\, \mathcal{H}_{\text{int}}(z)\right]\right\}|0\rangle}{\langle 0|T\exp\left[-i\int \mathrm{d}^4z\, \mathcal{H}_{\text{int}}(z)\right]|0\rangle}.$$

Here $|0\rangle$ is the free vacuum and $|\Omega\rangle$ is the interacting vacuum. The denominator removes disconnected vacuum factors. This formula is a central bridge between operator quantization and Feynman diagrams.

The fields inside the expectation value are free fields. Therefore Wick's theorem can reduce the time-ordered products to sums of propagators. This will be the subject of the next chapter.

7.8 Vacuum Persistence Amplitude

The vacuum persistence amplitude is

$$\langle 0|\hat{U}_I(\infty, -\infty)|0\rangle.$$

It is the amplitude for the free vacuum in the far past to become the free vacuum in the far future in the presence of interactions. Its perturbative expansion contains vacuum bubble diagrams: diagrams with no external legs.

These vacuum bubbles multiply correlation functions by an overall factor. In normalized expectation values, they cancel against the denominator. This is why connected diagrams, rather than all diagrams, carry the main physical content of scattering amplitudes.

7.9 Adiabatic Switching and In/Out States

Scattering theory compares states in the far past with states in the far future. The idealization is that interactions are effectively switched off at early and late times, so that asymptotic states can be described by free particles. This is expressed schematically by replacing the interaction with

$$\hat{H}_I(t) \longrightarrow e^{-\epsilon|t|}\hat{H}_I(t),$$

and taking $\epsilon \to 0^+$ at the end.

This adiabatic switching is a mathematical device. It helps define the connection between free Fock states and interacting scattering amplitudes. The states in the far past are called in-states, and those in the far future are called out-states.

The scattering operator is

$$\hat{S} = \hat{U}_I(\infty, -\infty) = T\exp\left(-i\int_{-\infty}^{\infty} \mathrm{d}t\, \hat{H}_I(t)\right).$$

Matrix elements of $\hat{S}$ between in- and out-states are the basic objects of perturbative scattering theory.

7.10 Why Exact Interacting QFT Is Difficult

The free theory is exactly solvable because it is a collection of independent oscillators. Interactions couple those oscillators. Once interactions are present, particle number may change, the vacuum may differ from the free vacuum, and exact operator solutions are usually unavailable.

Perturbation theory avoids solving the interacting theory exactly. It expands around the free theory and organizes corrections in powers of the coupling. This strategy is extremely powerful when the coupling is small and the relevant states are close to free asymptotic particle states.

However, perturbation theory has limits. Strong coupling, bound states, confinement, and nonperturbative vacuum effects require additional methods. The interaction picture is therefore not the whole of QFT, but it is the standard entry point into calculational quantum field theory.

Summary

The interaction picture separates the exactly solvable free Hamiltonian from the interaction Hamiltonian. Interaction-picture fields are free fields, while states evolve with $\hat{H}_I(t)$. The time evolution operator is a time-ordered exponential, and its expansion gives the Dyson series. This is the operator foundation of perturbation theory and the starting point for Wick's theorem and Feynman diagrams.

7.11 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish the Schroedinger, Heisenberg, and interaction pictures;
- explain why interaction-picture fields are free fields;
- write the interaction Hamiltonian for ϕ^4 theory and QED;
- define the interaction-picture time evolution operator;
- expand the time-ordered exponential as the Dyson series;
- explain perturbation theory as an expansion in couplings;

- describe the role of the vacuum persistence amplitude;
- explain why vacuum bubbles cancel in normalized correlators;
- define in-states, out-states, and the scattering operator;
- explain why exact interacting QFT is difficult.

Chapter 8

Wick Theorem and Diagrammatics

Remark

The Dyson series expresses interacting quantities as time-ordered products of free fields. Wick's theorem is the algebraic tool that evaluates those products. It reduces them to normal-ordered products plus contractions, and each contraction is a propagator.

8.1 Why Wick's Theorem Appears

In Chapter 7, perturbation theory led to expressions of the form

$$\langle 0|T\{\hat{\phi}_I(x_1)\cdots\hat{\phi}_I(x_n)\}|0\rangle.$$

The fields are free interaction-picture fields, but there may be many of them. To compute perturbative corrections, one needs a systematic way to evaluate such vacuum expectation values.

Wick's theorem provides exactly this. It says that a time-ordered product of free fields can be rewritten as a sum of normal-ordered products multiplied by contractions. In vacuum expectation values, the normal-ordered parts vanish unless all fields are contracted. Thus vacuum expectation values reduce to sums of products of propagators.

This is the algebraic origin of Feynman diagrams.

8.2 Normal Ordering Revisited

Normal ordering places all creation operators to the left of all annihilation operators. For a scalar field, we write

$$:\hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}}:=\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{k}}.$$

For fields, normal ordering means applying this rule after expanding the field into creation and annihilation parts. Schematically,

$$\hat{\phi}(x)=\hat{\phi}^{(+)}(x)+\hat{\phi}^{(-)}(x),$$

where $\hat{\phi}^{(+)}$ contains annihilation operators and $\hat{\phi}^{(-)}$ contains creation operators.

A normal-ordered product has zero vacuum expectation value unless it is just a number:

$$\langle 0|:\hat{\phi}(x_1)\cdots\hat{\phi}(x_n):|0\rangle=0$$

for $n>0$. This is why normal ordering is useful in perturbation theory.

8.3 Contractions

For two scalar fields, the contraction is defined as the difference between the time-ordered product and the normal-ordered product:

$$\mathrm{contr}(\hat{\phi}(x),\hat{\phi}(y))\equiv T\{\hat{\phi}(x)\hat{\phi}(y)\}-:\hat{\phi}(x)\hat{\phi}(y):.$$

For a free real scalar field, this contraction is the Feynman propagator:

$$\mathrm{contr}(\hat{\phi}(x), \hat{\phi}(y)) = D_F(x-y).$$

Equivalently,

$$\langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle = D_F(x-y).$$

A contraction is therefore not a new physical object. It is the propagator written as an algebraic operation inside a time-ordered product.

8.4 Wick Theorem for Scalar Fields

For free scalar fields, Wick's theorem states that

$$T\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} =: \hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n : + \text{all single contractions} + \text{all double contractions} + \cdots,$$

where $\hat{\phi}_i = \hat{\phi}(x_i)$. The expansion continues until all possible contractions have been included.

For two fields,

$$T\{\hat{\phi}_1\hat{\phi}_2\} =: \hat{\phi}_1\hat{\phi}_2 : + D_F(x_1-x_2).$$

For four fields,

$$\begin{aligned}\langle 0|T\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\}|0\rangle &= D_F(x_1-x_2)D_F(x_3-x_4)\\ &\quad + D_F(x_1-x_3)D_F(x_2-x_4)\\ &\quad + D_F(x_1-x_4)D_F(x_2-x_3).\end{aligned}$$

Only fully contracted terms survive between vacuum states.

Example

The vacuum expectation value of an odd number of free real scalar fields vanishes. There is no way to pair all fields into contractions.

8.5 Application to the Dyson Series

Consider ϕ^4 theory with

$$\mathcal{H}_{\mathrm{int}}(x) = \frac{\lambda}{4!}\hat{\phi}_I(x)^4.$$

The first-order correction to a two-point function contains

$$\frac{-i\lambda}{4!}\int \mathrm{d}^4z\, \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\hat{\phi}(z)^4\}|0\rangle.$$

Wick's theorem reduces this six-field expectation value to a sum of products of three propagators. Some contractions connect x and y to the interaction point z; others form vacuum bubbles or tadpole factors.

This example illustrates the general rule: interaction terms add field insertions, and Wick's theorem pairs those insertions into propagators.

8.6 Wick Theorem for Fermions

For fermionic fields, Wick's theorem has the same structure but includes signs. The basic fermion contraction is

$$\mathrm{contr}(\hat{\psi}(x), \hat{\bar{\psi}}(y)) = S_F(x-y),$$

where S_F is the Dirac propagator. Time ordering of fermions includes a minus sign when two fermionic operators are exchanged.

For example,

$$T\{\hat{\psi}(x)\hat{\bar{\psi}}(y)\} = \theta(x^0 - y^0)\hat{\psi}(x)\hat{\bar{\psi}}(y) - \theta(y^0 - x^0)\hat{\bar{\psi}}(y)\hat{\psi}(x).$$

The minus sign is not optional. It is required by the anticommutation relations of fermionic fields.

In diagrammatic language, every closed fermion loop contributes an additional minus sign. This sign has its origin in the permutations required to contract fermionic operators.

8.7 Fermionic Sign Factors

When applying Wick's theorem to fermions, one must count how many exchanges of fermionic operators are needed to bring the contracted fields next to each other. Each exchange contributes a minus sign.

A useful rule is:

$$\text{fermionic sign} = (-1)^{N_{\text{exchanges}}}.$$

For practical Feynman rules, these signs are packaged into standard rules: closed fermion loops give a minus sign, and external fermion lines must be ordered consistently with the chosen amplitude convention.

The sign bookkeeping is one of the main reasons diagrammatic notation is useful. It turns long operator manipulations into a structured set of graphical and algebraic rules.

8.8 From Algebra to Diagrams

Each term produced by Wick's theorem can be represented graphically. The rules are:

- each field insertion is represented by a point or an external leg;
- each contraction is represented by a line;
- each interaction Hamiltonian insertion is represented by a vertex;
- each vertex is integrated over spacetime;
- each line contributes the propagator of the corresponding field.

For ϕ^4 theory, each interaction vertex has four scalar lines attached to it. For QED, the interaction

$$e\bar{\psi}\gamma^\mu\psi A_\mu$$

has one photon line and two fermion lines attached to the vertex.

A Feynman diagram is therefore a picture of a term in the Wick expansion of the Dyson series. It is not a literal photograph of particles moving in spacetime.

8.9 External Lines, Internal Lines, and Vertices

External lines represent field insertions associated with initial or final states, or with operators whose correlation function is being computed. Internal lines represent contractions between fields inside the perturbative expansion. They carry propagators.

Vertices represent interaction terms. In ϕ^4 theory, the vertex factor is related to $-i\lambda$. In QED, the vertex factor is related to $-ie\gamma^\mu$, with conventions depending on the sign chosen for the interaction Lagrangian.

In momentum space, each internal line carries a momentum variable. Momentum is conserved at each vertex. Integrations over undetermined internal momenta become loop integrals.

8.10 Connected and Disconnected Diagrams

A diagram is connected if every part of it is linked to every other part by lines and vertices. A disconnected diagram factorizes into independent pieces.

In correlation functions, disconnected pieces often include vacuum bubbles, which are diagrams with no external lines. The normalized generating formulas of Chapter 7 divide by the vacuum persistence amplitude, causing pure vacuum bubbles to cancel.

Connected diagrams are the ones that carry the essential correlation between external insertions. Later, when we construct scattering amplitudes, connected amputated diagrams will play the central role.

8.11 Vacuum Bubbles

Vacuum bubbles arise from contractions among fields belonging only to interaction vertices, with no connection to external fields. For example, in ϕ^4 theory, a first-order vacuum bubble comes from contracting the four fields at a single interaction point among themselves.

Vacuum bubbles contribute to

$$\langle 0|T\exp\left[-i\int \mathrm{d}^4x\,\mathcal{H}_{\mathrm{int}}(x)\right]|0\rangle.$$

They affect the vacuum persistence amplitude but cancel from normalized correlation functions. This cancellation is why one often focuses on connected diagrams attached to the external points.

8.12 Symmetry Factors

The same diagram can arise from many equivalent Wick contractions. The symmetry factor accounts for this overcounting. In a perturbative expansion, the factors $1/n!$ from the Dyson series and factors such as $1/4!$ in the interaction Lagrangian combine with the number of contractions to produce the final coefficient of a diagram.

A symmetry factor is the inverse of the number of graph automorphisms that leave the diagram unchanged while preserving its external structure. In practice, one often learns symmetry factors by examples.

For instance, in ϕ^4 theory, the factor $1/4!$ is chosen so that the basic four-point vertex has a simple Feynman rule. Without this factorial, every vertex would carry extra combinatorial factors from permuting identical scalar fields.

8.13 Interpretation Limits of Feynman Diagrams

Feynman diagrams are extraordinarily useful, but they should be interpreted carefully. A diagram represents a term in a perturbative expansion, not a unique classical history. Internal lines do not represent directly observed particles. They represent propagators, or equivalently contractions of free fields.

This distinction matters especially for virtual particles. A virtual particle is not a particle detected in an experiment. It is a feature of a diagrammatic representation of a correlation function or amplitude. Different gauges or field variables can reorganize the same physical amplitude into different sets of intermediate diagrammatic terms.

The physical content lies in gauge-invariant amplitudes, cross sections, decay rates, and correlation functions, not in any single diagram by itself.

Summary

Wick's theorem converts time-ordered products of free fields into sums of normal-ordered products and contractions. Vacuum expectation values keep only fully contracted terms, and each contraction is a propagator. Applying Wick's theorem to the Dyson series produces Feynman diagrams: vertices come from interactions, internal lines from propagators, external lines from field insertions, and symmetry factors from combinatorics.

8.14 Chapter Checklist

After reading this chapter, one should be able to:

- explain why Wick's theorem is needed in perturbation theory;
- define normal ordering and contractions;
- state Wick's theorem for free scalar fields;
- evaluate a four-point free scalar vacuum expectation value;
- explain how contractions become propagators;
- state the fermionic version of Wick's theorem qualitatively;
- explain the origin of fermionic sign factors;
- translate Wick contractions into Feynman diagrams;
- distinguish external lines, internal lines, and vertices;
- explain connected diagrams, vacuum bubbles, and symmetry factors;
- state why Feynman diagrams should not be interpreted as literal particle trajectories.

Chapter 9

Scattering Theory and the S-Matrix

Remark

The previous chapters built the machinery of perturbative QFT: free fields, propagators, the interaction picture, Wick's theorem, and Feynman diagrams. Scattering theory explains how this machinery becomes physics. It connects correlation functions and diagrams to transition amplitudes, cross sections, and decay rates.

9.1 In-States and Out-States

Scattering theory compares what comes in from the far past with what goes out in the far future. The basic idealization is that particles are well separated at very early and very late times, so that they may be described by free Fock states.

An initial state is called an in-state and is denoted

$$|\alpha, \mathrm{in}\rangle.$$

A final state is called an out-state and is denoted

$$|\beta, \mathrm{out}\rangle.$$

For example, a two-particle initial state may be labelled by momenta and other quantum numbers,

$$|p_1, p_2, \mathrm{in}\rangle,$$

while a final state may contain different momenta or even a different number of particles.

The central physical question is: what is the amplitude for the initial state α to become the final state β?

9.2 The S-Matrix

The scattering matrix is defined by the overlap

$$S_{\beta\alpha} = \langle \beta, \mathrm{out} | \alpha, \mathrm{in}\rangle.$$

Equivalently, one introduces an operator $\hat{S}$ acting on free asymptotic states:

$$S_{\beta\alpha} = \langle \beta | \hat{S} | \alpha \rangle.$$

In the interaction picture,

$$\hat{S} = T \exp\left(-i \int_{-\infty}^{\infty} \mathrm{d}t\, \hat{H}_I(t)\right).$$

This is the same operator introduced at the end of Chapter 7. The novelty here is that its matrix elements are interpreted as scattering amplitudes.

It is useful to separate the identity contribution from the interacting part:

$$\hat{S} = 1 + i\hat{T}.$$

The identity term describes no scattering. The operator $\hat{T}$ contains the nontrivial transition information.

9.3 Transition Amplitudes

For relativistic scattering, momentum conservation is factored out. One writes

$$\langle\beta|i\hat{T}|\alpha\rangle = i(2\pi)^4\delta^{(4)}(P_\beta - P_\alpha)\mathcal{M}_{\beta\alpha}.$$

The quantity $\mathcal{M}_{\beta\alpha}$ is the invariant matrix element, or scattering amplitude. It is what Feynman rules are designed to compute.

The delta function expresses conservation of total four-momentum. The amplitude $\mathcal{M}$ contains the dynamical information: coupling constants, propagators, spinor factors, polarization vectors, and momentum dependence.

For example, in a tree-level scalar process, $\mathcal{M}$ may be a simple polynomial in the coupling. In QED it includes spinor bilinears and photon propagators. In loop calculations it includes integrals over internal momenta.

9.4 Matrix Elements and Momentum Conservation

Translation invariance implies four-momentum conservation. If the theory is invariant under spacetime translations, then the total four-momentum operator commutes with the scattering operator:

$$[\hat{P}^\mu, \hat{S}] = 0.$$

Therefore only transitions satisfying

$$P_\alpha^\mu = P_\beta^\mu$$

can have nonzero amplitude.

This is why every Feynman diagram in momentum space carries a momentum-conserving delta function at each vertex. After all internal vertex integrations are done, one overall delta function remains. It expresses total energy-momentum conservation for the full scattering process.

9.5 Lorentz-Invariant Phase Space

Probabilities are obtained by squaring amplitudes and summing or integrating over final states. The Lorentz-invariant one-particle phase-space measure is

$$\frac{\mathrm{d}^3 p}{(2\pi)^3}\frac{1}{2E_{\mathbf{p}}}.$$

For an n-particle final state with total incoming momentum P, the Lorentz-invariant phase space is

$$\mathrm{d}\Pi_n(P; p_1, \ldots, p_n) = (2\pi)^4\delta^{(4)}\left(P - \sum_{i=1}^{n} p_i\right)\prod_{i=1}^{n}\frac{\mathrm{d}^3 p_i}{(2\pi)^3 2E_i}.$$

This measure combines the on-shell conditions for final particles with total four-momentum conservation.

The same measure appears in cross sections and decay rates. It is one of the standard bridges between amplitudes and measurable predictions.

9.6 Cross Sections

For a two-particle scattering process

$$a + b \longrightarrow 1 + 2 + \cdots + n,$$

the differential cross section has the schematic form

$$d\sigma = \frac{1}{\text{flux}} |\mathcal{M}|^2 \, d\Pi_n.$$

The flux factor depends on the normalization of the incoming states. In a Lorentz-invariant form for two incoming particles with momenta p_a, p_b, it is built from

$$4\sqrt{(p_a \cdot p_b)^2 - m_a^2 m_b^2}.$$

Thus

$$d\sigma = \frac{|\mathcal{M}|^2}{4\sqrt{(p_a \cdot p_b)^2 - m_a^2 m_b^2}} \, d\Pi_n.$$

For massless or center-of-mass examples this expression simplifies, but the structure is always the same: amplitude squared times phase space divided by incoming flux.

9.7 Decay Rates

For a decay process

$$A \longrightarrow 1 + 2 + \cdots + n,$$

there is only one initial particle. The differential decay rate is

$$d\Gamma = \frac{1}{2M_A} |\mathcal{M}|^2 \, d\Pi_n,$$

where M_A is the mass of the decaying particle. The total decay rate is obtained by integrating over final-state phase space:

$$\Gamma = \int d\Gamma.$$

The lifetime is

$$\tau = \frac{1}{\Gamma}.$$

The same invariant amplitude technology therefore describes both scattering experiments and unstable-particle decays.

9.8 The LSZ Reduction Formula

Perturbation theory naturally computes time-ordered correlation functions. The LSZ reduction formula explains how to extract scattering amplitudes from those correlators.

For a scalar field, the rough idea is this: external particles correspond to poles of the Fourier-transformed correlation function. To obtain an amplitude, one multiplies by inverse propagator factors and takes the external momenta on shell. Schematically,

$$\mathcal{M} \sim \prod_i \left(p_i^2 - m^2\right) \widetilde{G}(p_1, \ldots, p_n)\Big|_{p_i^2 = m^2},$$

with normalization factors suppressed.

The exact LSZ formula includes wave-function renormalization factors and signs that depend on conventions. Its conceptual message is more important at this stage: scattering amplitudes are obtained from the pole structure of correlation functions.

9.9 From Correlation Functions to Scattering Amplitudes

The chain of ideas is now complete:

Lagrangian $\longrightarrow$ interaction Hamiltonian $\longrightarrow$ Dyson series $\longrightarrow$ Wick contractions $\longrightarrow$ Feynman diagrams $\longrightarrow$ $\mathcal{M}$

The first part of the chain is formal and operator-theoretic. The middle part is diagrammatic. The last part converts amplitudes into cross sections and decay rates.

This is why propagators and vertices are not merely mathematical decorations. They are the intermediate objects that connect a Lagrangian to measurable numbers.

9.10 Tree-Level $2 \to 2$ Scattering

A simple example is tree-level $2 \to 2$ scattering in ϕ^4 theory. The interaction is

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

At lowest order, there is one four-point vertex and no internal propagator. The amplitude is

$$i\mathcal{M} = -i\lambda, \qquad \mathcal{M} = -\lambda.$$

This example is deliberately simple. It shows that a contact interaction gives a constant tree-level amplitude. More complicated theories produce amplitudes with propagators, spinor factors, and polarization vectors.

In momentum space, the full matrix element is

$$\langle p_3, p_4 | i\hat{T} | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4)\mathcal{M}.$$

The delta function enforces momentum conservation; $\mathcal{M}$ is the part used in the cross-section formula.

9.11 Decay of an Unstable Particle

Suppose a particle A can decay into two particles 1 and 2. If the amplitude is $\mathcal{M}(A \to 1 + 2)$, then

$$\mathrm{d}\Gamma = \frac{1}{2M_A}|\mathcal{M}|^2(2\pi)^4\delta^{(4)}(p_A - p_1 - p_2)\frac{\mathrm{d}^3p_1}{(2\pi)^3 2E_1}\frac{\mathrm{d}^3p_2}{(2\pi)^3 2E_2}.$$

This formula is the decay analogue of the scattering cross section. It says that rates are determined by amplitude squared times available phase space.

The phase space can strongly affect the physical rate. A large amplitude may produce a small rate if very little phase space is available, while a modest amplitude can produce a large rate if many final states are accessible.

9.12 From Amplitudes to Measurable Predictions

Experiments do not measure Feynman diagrams directly. They measure cross sections, angular distributions, branching ratios, decay widths, and event rates. QFT predicts these quantities by computing amplitudes and integrating them over the appropriate phase space.

A realistic prediction usually requires additional steps: spin sums, averages over initial polarizations, sums over final polarizations, symmetry factors for identical particles, detector cuts, and sometimes radiative corrections. The basic structure, however, is already visible:

$$\text{theory} \longrightarrow \mathcal{M} \longrightarrow |\mathcal{M}|^2 \longrightarrow \text{phase-space integral} \longrightarrow \text{observable}.$$

Summary

Scattering theory turns the operator and diagrammatic machinery of perturbative QFT into measurable predictions. The S-matrix connects in-states to out-states. Momentum conservation is factored out to define the invariant amplitude $\mathcal{M}$. Cross sections and decay rates are obtained from $|\mathcal{M}|^2$ integrated over Lorentz-invariant phase space. LSZ explains why scattering amplitudes are encoded in the pole structure of time-ordered correlation functions.

9.13 Chapter Checklist

After reading this chapter, one should be able to:

- define in-states, out-states, and the S-matrix;
- distinguish S, T, and the invariant amplitude $\mathcal{M}$;
- explain the origin of the momentum-conserving delta function;
- write the Lorentz-invariant phase-space measure;
- state the schematic cross-section formula;
- state the decay-rate formula;
- explain the conceptual role of LSZ reduction;
- describe how correlation functions lead to amplitudes;
- compute the tree-level ϕ^4 amplitude at lowest order;
- explain how amplitudes become measurable predictions.

Chapter 10

Path Integral for Quantum Mechanics

Remark

The operator formalism led from fields to propagators, diagrams, and scattering amplitudes. The path integral gives a different but equivalent viewpoint: a transition amplitude is computed as a sum over histories, weighted by e^{iS}. We first develop this idea in ordinary quantum mechanics before extending it to fields.

10.1 Transition Amplitudes as Sums Over Histories

In the operator formulation of quantum mechanics, the transition amplitude from position q_i at time t_i to position q_f at time t_f is

$$\langle q_f, t_f | q_i, t_i \rangle = \langle q_f | e^{-i\hat{H}(t_f - t_i)} | q_i \rangle.$$

The path-integral idea is that the same amplitude can be written as a sum over all paths connecting the endpoints:

$$\langle q_f, t_f | q_i, t_i \rangle = \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}q(t)\, e^{iS[q]}.$$

Here

$$S[q] = \int_{t_i}^{t_f} \mathrm{d}t\, L(q, \dot{q})$$

is the classical action evaluated on the path $q(t)$. The notation $\mathcal{D}q(t)$ means integration over all intermediate paths.

This formula should not be interpreted as saying that the particle secretly chooses one path. Rather, the transition amplitude is built by coherently summing contributions from all paths.

10.2 Time Slicing

The path integral can be derived by dividing the time interval into many small steps. Let

$$t_f - t_i = N\epsilon.$$

Insert resolutions of the identity in the position basis:

$$1 = \int \mathrm{d}q_j\, |q_j\rangle\langle q_j|.$$

Then

$$\langle q_f | e^{-i\hat{H}(t_f - t_i)} | q_i \rangle = \int \prod_{j=1}^{N-1} \mathrm{d}q_j \prod_{j=0}^{N-1} \langle q_{j+1} | e^{-i\epsilon\hat{H}} | q_j \rangle.$$

For a Hamiltonian

$$H = \frac{p^2}{2m} + V(q),$$

one also inserts momentum states and evaluates the short-time matrix element. In the limit $N \to \infty$, this produces

$$\int \mathcal{D}q \, e^{i\int \mathrm{d}t\,[\frac{1}{2}m\dot{q}^2 - V(q)]}.$$

Thus the path integral is not a new postulate here; it is a reorganization of operator time evolution.

10.3 Classical Limit and Stationary Phase

The factor $e^{iS[q]}$ oscillates rapidly when the action is large compared with $\hbar$. Restoring $\hbar$, the weight is

$$e^{iS[q]/\hbar}.$$

In the limit $\hbar \to 0$, most paths interfere destructively. The dominant contribution comes from paths for which the action is stationary:

$$\delta S = 0.$$

This condition gives the classical equations of motion.

Thus the classical path is not the only path included in the quantum amplitude. It is the path around which neighbouring contributions add coherently in the classical limit. Quantum mechanics is the sum over all histories; classical mechanics is the stationary-phase approximation.

10.4 Gaussian Integrals

Gaussian integrals are the technical backbone of free path integrals. The ordinary one-dimensional formula is

$$\int_{-\infty}^{\infty} \mathrm{d}x \, e^{-\frac{1}{2}ax^2 + Jx} = \sqrt{\frac{2\pi}{a}} \, e^{\frac{J^2}{2a}},$$

for $a > 0$. In oscillatory form one uses an $i\epsilon$ prescription to make the integral well defined.

The multidimensional version is

$$\int \mathrm{d}^n x \, \exp\left[-\frac{1}{2}x^T A x + J^T x\right] = \left(\frac{(2\pi)^n}{\det A}\right)^{1/2} \exp\left(\frac{1}{2}J^T A^{-1} J\right).$$

Free quantum theories lead to infinite-dimensional versions of this formula. The inverse operator A^{-1} becomes the propagator.

10.5 The Free Particle

For a free particle,

$$L = \frac{1}{2}m\dot{q}^2.$$

The path integral is Gaussian. The result is

$$K(q_f, t_f; q_i, t_i) = \left(\frac{m}{2\pi i T}\right)^{1/2} \exp\left[\frac{im(q_f - q_i)^2}{2T}\right],$$

where $T = t_f - t_i$. The exponent is the classical action evaluated on the straight-line path connecting the endpoints.

This example illustrates a common pattern. For a quadratic action, the path integral can be evaluated exactly. The answer is a normalization determinant multiplying $e^{iS_{\mathrm{cl}}}$.

10.6 The Harmonic Oscillator

For the harmonic oscillator,

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2.$$

The action is still quadratic, so the path integral remains Gaussian. The transition amplitude again has the form

$$K(q_f, t_f; q_i, t_i) = \mathcal{N}(T)e^{iS_{\text{cl}}(q_f,t_f;q_i,t_i)}.$$

The exact normalization $\mathcal{N}(T)$ comes from the determinant of the quadratic fluctuation operator.

The oscillator is important because a free field is a collection of oscillators. The free-field path integral will therefore be an infinite product of Gaussian oscillator path integrals.

10.7 Generating Functionals in Quantum Mechanics

To compute correlation functions, introduce a source $J(t)$ coupled to $q(t)$:

$$Z[J] = \int \mathcal{D}q \, \exp\left(iS[q] + i\int \mathrm{d}t\, J(t)q(t)\right).$$

The source is an auxiliary function. It is set to zero after differentiating. Correlation functions are obtained by functional derivatives:

$$\langle T q(t_1)q(t_2)\rangle = \frac{1}{Z[0]}\,\frac{1}{i^2}\frac{\delta^2 Z[J]}{\delta J(t_1)\delta J(t_2)}\bigg|_{J=0}.$$

This is the prototype of the generating functional method used in QFT.

10.8 Sources and Correlation Functions

The source term keeps track of insertions. Differentiating once gives

$$\frac{1}{i}\frac{\delta Z[J]}{\delta J(t)} = \int \mathcal{D}q\, q(t)\exp\left(iS[q] + i\int \mathrm{d}t'\, J(t')q(t')\right).$$

Differentiating twice inserts two factors of q, and so on. Therefore

$$\frac{1}{i^n}\frac{\delta^n Z[J]}{\delta J(t_1)\cdots\delta J(t_n)}\bigg|_{J=0}$$

generates the n-point time-ordered correlation function.

In operator language these are vacuum expectation values of time-ordered products. In the path-integral language they are moments of a functional integral with an oscillatory weight.

10.9 Why the Path Integral Is Useful for QFT

The path integral is especially useful in field theory for several reasons. First, it makes spacetime symmetries more visible because the action appears directly. Second, it gives a compact way to derive Feynman rules. Third, it handles correlation functions and sources naturally. Fourth, it generalizes to gauge fixing, ghosts, effective actions, and nonperturbative approximations.

For a field, the single coordinate $q(t)$ is replaced by a field configuration $\phi(x)$. The ordinary source $J(t)$ is replaced by a spacetime source $J(x)$. The path integral becomes

$$\int \mathcal{D}\phi\, e^{iS[\phi]}.$$

This is the subject of the next chapter.

10.10 Relation to Canonical Quantization

The path integral and canonical quantization are not separate quantum theories. They are two formulations of the same theory. Canonical quantization emphasizes Hilbert spaces, operators, commutators, and states. The path integral emphasizes action functionals, histories, sources, and correlation functions.

The connection can be summarized as

$$\langle q_f | e^{-i\hat{H}T} | q_i \rangle = \int_{q_i}^{q_f} \mathcal{D}q \, e^{iS[q]}.$$

The left-hand side is canonical. The right-hand side is the path integral. In field theory, this equivalence underlies the use of functional integrals to compute the same time-ordered correlators that appear in the operator formalism.

Summary

The path integral expresses a transition amplitude as a sum over histories weighted by e^{iS}. Time slicing connects this formula to operator evolution. The classical equations arise by stationary phase. Quadratic actions lead to Gaussian integrals, whose inverses become propagators. Adding sources produces a generating functional whose derivatives give time-ordered correlation functions.

10.11 Chapter Checklist

After reading this chapter, one should be able to:

- write a transition amplitude as a path integral over histories;
- explain the time-slicing derivation;
- explain how the classical limit follows from stationary phase;
- evaluate the role of Gaussian integrals in free theories;
- state the path integral for the free particle and harmonic oscillator;
- define a generating functional with a source;
- obtain correlation functions by differentiating with respect to the source;
- explain why the path integral is useful for QFT;
- relate the path integral to canonical quantization.

Chapter 11

Path Integral for Scalar Fields

Remark

The path integral for a particle sums over histories $q(t)$. The path integral for a field sums over field configurations $\phi(x)$. This chapter develops the functional integral for scalar fields and shows how propagators, correlation functions, perturbation theory, and diagrams arise from a single generating functional.

11.1 From Paths to Field Configurations

In ordinary quantum mechanics, the path integral is

$$\int \mathcal{D}q(t)\, e^{iS[q]}.$$

For a scalar field, the dynamical variable is not a single function of time but a function of spacetime. The corresponding path integral is

$$\int \mathcal{D}\phi\, e^{iS[\phi]}.$$

The measure $\mathcal{D}\phi$ symbolically means integration over all field configurations.

For a real scalar field, the action is

$$S[\phi] = \int \mathrm{d}^4x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2\right)$$

for the free theory. Interactions are added by including non-quadratic terms, for example

$$S[\phi] = \int \mathrm{d}^4x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right).$$

11.2 The Generating Functional

Introduce a source $J(x)$ coupled linearly to the field:

$$Z[J] = \int \mathcal{D}\phi \,\exp\left(iS[\phi] + i\int \mathrm{d}^4x\, J(x)\phi(x)\right).$$

The source is an auxiliary function. It is used to generate correlation functions and then set to zero.

The vacuum-to-vacuum functional is

$$Z[0] = \int \mathcal{D}\phi\, e^{iS[\phi]}.$$

Normalized correlation functions are obtained by dividing by $Z[0]$. For example,

$$\langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \frac{1}{Z[0]}\,\frac{1}{i^2}\,\frac{\delta^2 Z[J]}{\delta J(x)\delta J(y)}\bigg|_{J=0}.$$

Thus the path integral generates the same time-ordered correlators that appeared in the operator formulation.

11.3 The Free Scalar Generating Functional

The free scalar action can be written, after integrating by parts and ignoring boundary terms, as

$$S_0[\phi] = -\frac{1}{2}\int \mathrm{d}^4x\, \phi(x)(\Box + m^2)\phi(x).$$

With a source, the exponent is quadratic in ϕ:

$$iS_0[\phi] + i\int \mathrm{d}^4x\, J\phi.$$

This is an infinite-dimensional Gaussian integral. The result has the form

$$Z_0[J] = Z_0[0]\exp\left(\frac{i}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right),$$

where D_F is the Feynman propagator satisfying

$$(\Box_x + m^2)D_F(x-y) = -i\delta^{(4)}(x-y).$$

The inverse of the quadratic operator in the action is therefore the propagator. This is the path-integral version of the result obtained from canonical quantization.

11.4 Sources and Functional Derivatives

Functional derivatives insert fields. The basic rule is

$$\frac{1}{i}\frac{\delta}{\delta J(x)}Z[J] = \int \mathcal{D}\phi\,\phi(x)\exp\left(iS[\phi] + i\int \mathrm{d}^4z\, J(z)\phi(z)\right).$$

Therefore

$$\langle T\phi(x_1)\cdots\phi(x_n)\rangle = \frac{1}{Z[0]}\,\frac{1}{i^n}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}.$$

For the free theory, differentiating $Z_0[J]$ reproduces Wick's theorem. The exponential quadratic in J generates all pairings of the external points, and each pair contributes one propagator.

11.5 Gaussian Functional Integrals

The free scalar generating functional is the infinite-dimensional analogue of

$$\int \mathrm{d}^nx\,\exp\left[-\frac{1}{2}x^TAx + J^Tx\right] \propto \exp\left(\frac{1}{2}J^TA^{-1}J\right).$$

In field theory, the matrix A becomes a differential operator, and A^{-1} becomes a Green function.

Schematically,

$$A = \Box + m^2, \qquad A^{-1} \sim D_F.$$

The $i\epsilon$ prescription specifies which inverse is meant. Without it, the operator has several possible Green functions. With it, the inverse is the Feynman propagator appropriate for time-ordered vacuum correlators.

The determinant of the quadratic operator contributes to $Z_0[0]$. In many normalized correlation functions this factor cancels, but determinants become important in loop effects, effective actions, and vacuum energies.

11.6 Interacting Scalar Theory

For ϕ^4 theory,

$$S[\phi] = S_0[\phi] - \int \mathrm{d}^4x \, \frac{\lambda}{4!} \phi(x)^4.$$

The generating functional is

$$Z[J] = \int \mathcal{D}\phi \, \exp\left(iS_0[\phi] - i \int \mathrm{d}^4x \, \frac{\lambda}{4!} \phi^4 + i \int \mathrm{d}^4x \, J\phi \right).$$

The interaction can be represented as functional derivatives acting on the free generating functional:

$$Z[J] = \exp\left[-i \int \mathrm{d}^4x \, \frac{\lambda}{4!} \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^4 \right] Z_0[J].$$

Expanding the exponential in powers of λ gives perturbation theory. This is the path-integral version of the Dyson expansion.

11.7 Diagrammatic Expansion

The diagrammatic rules follow from expanding the interacting generating functional and differentiating with respect to sources. Each factor of $D_F(x-y)$ becomes a line. Each factor of $-i\lambda \int \mathrm{d}^4x$ becomes an interaction vertex integrated over spacetime.

For ϕ^4 theory:

- a propagator contributes $D_F(x-y)$ in position space or $i/(k^2 - m^2 + i\epsilon)$ in momentum space;
- a vertex contributes a factor $-i\lambda$;
- four scalar lines meet at each vertex;
- one integrates over internal vertex positions or internal momenta;
- symmetry factors account for equivalent contractions.

Thus the Feynman rules derived earlier from Wick's theorem also emerge directly from the functional integral.

11.8 Connected Generating Functional

The generating functional $Z[J]$ includes disconnected diagrams. It is useful to define the connected generating functional $W[J]$ by

$$Z[J] = e^{iW[J]}.$$

Equivalently,

$$W[J] = -i \ln Z[J].$$

Functional derivatives of $W[J]$ generate connected correlation functions. For example,

$$\frac{1}{i} \frac{\delta W[J]}{\delta J(x)}$$

gives the connected one-point function, and higher derivatives give connected higher-point functions.

The logarithm removes the disconnected multiplication structure of $Z[J]$. This is the functional-integral version of the statement that disconnected vacuum bubbles exponentiate and cancel in normalized correlators.

11.9 The Classical Field

The source-dependent expectation value of the field is defined by

$$\phi_c(x) = \frac{\delta W[J]}{\delta J(x)}.$$

It is called the classical field, although it is really the expectation value of the quantum field in the presence of a source. When the source is removed, ϕ_c may vanish or may describe a nontrivial vacuum expectation value.

This quantity is important because it is the natural variable of the effective action. Instead of describing the theory as a functional of the external source J, one can describe it as a functional of the expectation value ϕ_c.

11.10 The Effective Action

The effective action $\Gamma[\phi_c]$ is defined by a Legendre transform:

$$\Gamma[\phi_c] = W[J] - \int \mathrm{d}^4x\, J(x)\phi_c(x),$$

where J is understood as a functional of ϕ_c. It satisfies

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x).$$

When the source is set to zero, the quantum equation of motion is

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = 0.$$

At tree level, Γ is just the classical action. Loop corrections modify it. Therefore the effective action is the quantum-corrected action governing expectation values.

11.11 One-Particle-Irreducible Functions

Functional derivatives of $\Gamma[\phi_c]$ generate one-particle-irreducible correlation functions, abbreviated 1PI functions. A diagram is one-particle irreducible if it cannot be disconnected by cutting a single internal line.

The two-point 1PI function is related to the inverse full propagator. The four-point 1PI function is related to the quantum-corrected interaction vertex. This language becomes essential in renormalization, where masses, fields, and couplings are corrected by loop effects.

The hierarchy is therefore

$$Z[J] \longrightarrow \text{all diagrams}, \qquad W[J] \longrightarrow \text{connected diagrams}, \qquad \Gamma[\phi_c] \longrightarrow \text{1PI diagrams}.$$

11.12 Classical Limit and Saddle Points

The path integral also gives a useful view of the classical limit. When the action is large compared with $\hbar$, the integral is dominated by stationary points of the action:

$$\frac{\delta S}{\delta\phi(x)} = 0.$$

This is the classical field equation. Quantum corrections come from fluctuations around the saddle point.

For a free theory, fluctuations are Gaussian. For an interacting theory, fluctuations generate vertices and loop diagrams. Thus perturbation theory can be viewed as a systematic expansion around a saddle point of the action.

Summary

The scalar-field path integral sums over field configurations rather than particle paths. The source functional $Z[J]$ generates time-ordered correlation functions. For a free scalar field, the functional integral is Gaussian and the inverse quadratic operator is the Feynman propagator. Interactions are generated by functional derivatives and produce the same diagrammatic expansion as Wick's theorem. The functionals $W[J]$ and $\Gamma[\phi_c]$ organize connected and one-particle-irreducible diagrams.

11.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain the meaning of the functional measure $\mathcal{D}\phi$;
- write the scalar-field generating functional with a source;
- derive the free generating functional $Z_0[J]$;
- obtain correlation functions by differentiating with respect to J;
- explain why the free path integral is Gaussian;
- express interactions as functional derivative operators;
- derive the diagrammatic expansion from $Z[J]$;
- define $W[J]$ and explain connected diagrams;
- define the classical field ϕ_c;
- define the effective action $\Gamma[\phi_c]$;
- explain the meaning of one-particle-irreducible functions.

Chapter 12

Fermionic Path Integrals

Remark

Scalar path integrals integrate over ordinary commuting fields. Fermionic path integrals require anticommuting variables. The price is algebraic unfamiliarity; the reward is a compact functional-integral description of Dirac fields, fermion propagators, fermion loops, and determinants.

12.1 Why Fermions Need a Different Integral

The scalar path integral sums over ordinary field configurations:

$$Z[J] = \int \mathcal{D}\phi \, e^{iS[\phi]+i\int J\phi}.$$

This works because scalar fields are bosonic. Their quantum operators obey commutation relations, and the corresponding functional variables commute.

Dirac fields are different. Their operators obey anticommutation relations, and fermionic many-particle states are antisymmetric. A path integral for fermions must reproduce this algebra. The appropriate variables are therefore not ordinary complex numbers, but Grassmann variables: objects that anticommute.

The transition is

$$\text{bosons} \longrightarrow \text{commuting variables}, \qquad \text{fermions} \longrightarrow \text{anticommuting variables}.$$

This chapter develops the basic calculus needed for fermionic functional integrals.

12.2 Grassmann Variables

A Grassmann variable θ anticommutes with another Grassmann variable χ:

$$\theta\chi = -\chi\theta.$$

In particular,

$$\theta^2 = 0.$$

This nilpotency makes every function of a single Grassmann variable finite:

$$f(\theta) = a + b\theta.$$

For several Grassmann variables, polynomials terminate because each variable can appear at most once.

Grassmann variables also anticommute with fermionic sources and fermionic fields, but commute with ordinary numbers and bosonic variables. They are bookkeeping objects that encode fermionic antisymmetry inside a functional integral.

12.3 Berezin Integration

Integration over Grassmann variables is called Berezin integration. It is defined by the rules

$$\int \mathrm{d}\theta\, 1 = 0, \qquad \int \mathrm{d}\theta\, \theta = 1.$$

Thus Grassmann integration behaves like differentiation. For a function $f(\theta) = a + b\theta$,

$$\int \mathrm{d}\theta\, f(\theta) = b.$$

For two variables, the order matters:

$$\int \mathrm{d}\theta_2\, \mathrm{d}\theta_1\, \theta_1\theta_2 = 1.$$

Changing the order of variables or integration measures can introduce minus signs. This is the algebraic origin of many sign factors in fermionic path integrals.

12.4 Gaussian Integrals for Grassmann Variables

For one pair of independent Grassmann variables $\bar{\theta}, \theta$,

$$\int \mathrm{d}\bar{\theta}\, \mathrm{d}\theta\, e^{-\bar{\theta}a\theta} = a.$$

For many variables, the general formula is

$$\int \prod_i \mathrm{d}\bar{\theta}_i\, \mathrm{d}\theta_i\, \exp\left(-\bar{\theta}_i A_{ij} \theta_j\right) = \det A.$$

This differs from the bosonic Gaussian integral, which produces $(\det A)^{-1/2}$ for real variables or $(\det A)^{-1}$ for complex commuting variables. Fermions produce determinants in the numerator.

With sources, one has

$$\int \mathrm{d}\bar{\theta}\, \mathrm{d}\theta\, \exp\left(-\bar{\theta}A\theta + \bar{\eta}\theta + \bar{\theta}\eta\right) = \det A\, \exp\left(\bar{\eta}A^{-1}\eta\right),$$

up to conventions for signs and ordering. The inverse matrix A^{-1} will become the fermion propagator.

12.5 The Dirac Generating Functional

The free Dirac action is

$$S_0[\bar{\psi}, \psi] = \int \mathrm{d}^4x\, \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

In the path integral, ψ and $\bar{\psi}$ are independent Grassmann-valued fields. Introduce Grassmann-valued sources η and $\bar{\eta}$:

$$Z_0[\bar{\eta}, \eta] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, \exp\left(iS_0[\bar{\psi}, \psi] + i\int \mathrm{d}^4x\, [\bar{\eta}\psi + \bar{\psi}\eta]\right).$$

This is a Gaussian Grassmann functional integral. It gives

$$Z_0[\bar{\eta}, \eta] = Z_0[0, 0]\exp\left(-i\int \mathrm{d}^4x\, \mathrm{d}^4y\, \bar{\eta}(x)S_F(x-y)\eta(y)\right),$$

where S_F is the Dirac Feynman propagator.

The sign in the exponent depends on source-ordering conventions. The essential point is invariant: differentiating with respect to fermionic sources produces Dirac propagators and fermionic correlation functions.

12.6 Dirac Propagator from the Path Integral

The Dirac propagator is the inverse of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m)S_F(x-y) = i\delta^{(4)}(x-y).$$

In momentum space,

$$S_F(p) = \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon}.$$

From the generating functional,

$$\langle 0|T\hat{\psi}_\alpha(x)\hat{\bar{\psi}}_\beta(y)|0\rangle = S_{F,\alpha\beta}(x-y).$$

Thus the same propagator obtained by canonical quantization appears as the inverse operator in the fermionic Gaussian integral.

The path-integral perspective makes the parallel with scalar fields clear:

$$\text{quadratic operator in the action}^{-1} = \text{free propagator}.$$

For scalars the operator is Klein–Gordon. For fermions it is Dirac.

12.7 Generating Fermionic Correlation Functions

Fermionic correlation functions are generated by differentiating with respect to Grassmann sources. Schematically,

$$\langle T\psi(x_1)\bar{\psi}(y_1)\cdots\psi(x_n)\bar{\psi}(y_n)\rangle$$

is obtained by applying derivatives with respect to $\bar{\eta}(x_i)$ and $\eta(y_i)$ to $Z[\bar{\eta},\eta]$, then setting the sources to zero.

The order of the derivatives matters. Since the sources anticommute, exchanging two source derivatives can produce a minus sign. These signs are the path-integral version of the fermionic signs in Wick's theorem.

A useful practical rule is that the path integral automatically keeps track of fermionic antisymmetry if the source ordering is used consistently.

12.8 Interactions with Fermions

For QED, the interaction term is

$$S_{\text{int}} = -e\int \mathrm{d}^4x\, \bar{\psi}\gamma^\mu\psi A_\mu.$$

The full generating functional is

$$Z[\bar{\eta},\eta,J] = \int \mathcal{D}A\mathcal{D}\bar{\psi}\mathcal{D}\psi\ \exp\left(iS[A,\bar{\psi},\psi] + i\int \mathrm{d}^4x\,[J_\mu A^\mu + \bar{\eta}\psi + \bar{\psi}\eta]\right).$$

Perturbation theory is obtained by expanding the interaction exponential or by rewriting interactions as functional derivatives with respect to sources.

The QED vertex contains one photon field, one ψ, and one $\bar{\psi}$. Therefore each vertex joins one photon line and two fermion lines. This is the path-integral origin of the usual QED diagrammatic vertex.

12.9 Fermion Loops and Minus Signs

Closed fermion loops carry an extra minus sign. In the operator formalism this comes from permuting fermionic operators in Wick contractions. In the path integral it comes from the algebra of Grassmann variables and the determinant produced by integrating out fermions.

A closed fermion loop is associated with a trace over spinor indices and an integral over loop momentum. The additional sign is part of the Feynman rule:

$$\text{each closed fermion loop contributes a factor } -1.$$

This rule is not an arbitrary convention. It is a direct consequence of fermionic anticommutation.

12.10 Determinants from Integrating Out Fermions

Because the fermionic path integral is Gaussian when the action is quadratic in $\psi, \bar{\psi}$, one can formally integrate out fermions. For a Dirac operator D,

$$\int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, e^{i\int \bar{\psi}D\psi} \propto \det D.$$

If the Dirac operator depends on a background gauge field, the determinant also depends on that background:

$$\det(i\gamma^\mu D_\mu - m).$$

This determinant encodes the effect of virtual fermion loops on the gauge-field dynamics.

For bosonic Gaussian integrals, one typically gets inverse determinants. This difference between bosonic and fermionic determinants is one of the key structural distinctions between commuting and anticommuting fields.

12.11 Majorana Fermions in the Path Integral

A Majorana fermion is constrained to be related to its charge conjugate. In the path integral this means that the independent fermionic variables are reduced relative to a Dirac field. The Gaussian integral no longer gives an ordinary determinant in the same way; it gives a Pfaffian.

Schematically, for an antisymmetric operator A,

$$\int \mathcal{D}\chi\, e^{-\frac{1}{2}\chi A\chi} \propto \mathrm{Pf}(A),$$

and

$$\mathrm{Pf}(A)^2 = \det A.$$

The details require a careful treatment of charge conjugation and spinor conventions. At this stage, the main lesson is that Majorana path integrals are fermionic Gaussian integrals with fewer independent variables than Dirac path integrals.

12.12 Physical Interpretation and Common Pitfalls

Grassmann variables are not classical values of a fermion field. They are anticommuting integration variables designed to reproduce fermionic quantum statistics. One should not imagine a fermionic path integral as a sum over ordinary classical spinor configurations.

Another common pitfall is to treat ψ and $\bar{\psi}$ as complex conjugates inside the functional integral. In the fermionic path integral they are independent Grassmann variables. Their relation as adjoint operators belongs to the canonical operator formulation, while the functional integral uses an independent-variable representation.

Finally, fermionic determinants and loop signs are not optional decorations. They are the mechanism by which the Pauli principle and virtual fermion effects enter the path-integral formalism.

Summary

Fermionic path integrals use Grassmann variables. Berezin integration turns fermionic Gaussian integrals into determinants, and source derivatives generate fermionic time-ordered correlation functions. The inverse Dirac operator is the Dirac propagator. Fermion loops acquire minus signs, and integrating out fermions produces determinants that encode virtual fermion effects. Majorana fermions require a related Pfaffian structure.

12.13 Chapter Checklist

After reading this chapter, one should be able to:

- define Grassmann variables and state their anticommutation property;
- state the basic rules of Berezin integration;
- evaluate the structure of a Gaussian Grassmann integral;
- write the free Dirac generating functional with Grassmann sources;
- identify the Dirac propagator as the inverse Dirac operator;
- explain how fermionic correlation functions are generated by source derivatives;
- explain why closed fermion loops carry a minus sign;
- describe how integrating out fermions produces determinants;
- explain the basic path-integral difference between Dirac and Majorana fermions;
- avoid interpreting Grassmann fields as ordinary classical spinor configurations.

Chapter 13

Gauge Fields in the Path Integral

Remark

Gauge fields require special care in the path integral because the functional integral over A_μ overcounts physically equivalent configurations. Gauge fixing removes this overcounting. In non-Abelian theories, the gauge-fixing procedure introduces ghost fields and leads naturally to BRST symmetry.

13.1 Overcounting Gauge-Equivalent Configurations

For electromagnetism, the gauge potential transforms as

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda.$$

The two potentials describe the same physical electromagnetic field. Therefore a naive path integral

$$\int \mathcal{D}A_\mu \, e^{iS[A]}$$

integrates over infinitely many copies of the same physical configuration.

This is not merely an aesthetic problem. The redundancy makes the quadratic operator non-invertible, so the propagator of A_μ is not defined until one chooses a gauge. In operator language this was the statement that not all components of A_μ are physical. In the path-integral language it becomes a problem of dividing by the volume of the gauge group.

13.2 Gauge Fixing in the Functional Integral

A gauge condition is a function $G[A]$ imposed on gauge fields. A common covariant choice is

$$G[A] = \partial_\mu A^\mu.$$

Instead of integrating over all representatives of each gauge orbit, one wants to integrate once over each physical configuration. Formally,

$$\int \mathcal{D}A_\mu \quad \longrightarrow \quad \int \mathcal{D}A_\mu \, \delta(G[A]) \, \Delta[A],$$

where $\Delta[A]$ is a determinant that compensates for the change of variables from gauge orbits to gauge-fixed representatives.

In practice, one often replaces the sharp delta-function condition by a Gaussian average over gauges. This leads to the covariant gauge-fixing term

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The parameter ξ labels the gauge. The choice $\xi = 1$ is Feynman gauge.

13.3 The Faddeev–Popov Method

The Faddeev–Popov method inserts a carefully chosen form of unity into the path integral:

$$1 = \Delta_{\mathrm{FP}}[A] \int \mathcal{D}\alpha\, \delta(G[A^\alpha]).$$

Here A^α denotes the gauge transform of A, and $\Delta_{\mathrm{FP}}[A]$ is the Faddeev–Popov determinant.

The determinant is given formally by

$$\Delta_{\mathrm{FP}}[A] = \det\left(\frac{\delta G[A^\alpha]}{\delta\alpha}\right).$$

This factor tells us how the gauge condition changes as one moves along a gauge orbit. It is the Jacobian associated with gauge fixing.

For Abelian gauge theory in a linear covariant gauge, this determinant is field-independent. It contributes only an overall constant and is usually ignored. For non-Abelian gauge theory, the determinant depends on the gauge field and has dynamical consequences.

13.4 Ghost Fields

A determinant can be represented as a Grassmann integral. For an operator M,

$$\det M = \int \mathcal{D}\bar{c}\mathcal{D}c\, \exp\left(i\int \mathrm{d}^4x\, \bar{c}Mc\right),$$

with appropriate conventions. The Grassmann fields c and $\bar{c}$ are called ghost fields.

Ghosts are anticommuting scalar fields. This phrase sounds paradoxical: they have scalar Lorentz transformation properties but fermionic statistics. They are not introduced because the theory has new physical particles. They are introduced because the gauge-fixing determinant must be represented inside the functional integral.

For non-Abelian gauge theory, ghosts interact with gauge fields. Their diagrams are essential for unitarity and gauge invariance in covariant gauges.

13.5 Why Ghosts Are Not Physical Particles

Ghost fields appear as internal lines in Feynman diagrams, but they do not appear as physical external states. They are artifacts of gauge fixing, not quanta in the physical Hilbert space.

This is analogous to the unphysical photon polarizations discussed in Chapter 5. Covariant quantization introduces redundant components for calculational convenience. The formalism must then ensure that unphysical states do not appear in physical observables.

Ghosts are part of that mechanism. Their contributions cancel gauge-dependent and unphysical pieces of gauge-field loops. Removing them would spoil the consistency of non-Abelian perturbation theory.

13.6 Gauge-Fixed Photon Path Integral

For QED, the gauge-fixed action is

$$S[A] = \int \mathrm{d}^4x\left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2\right].$$

The corresponding generating functional is

$$Z[J] = \int \mathcal{D}A\, \exp\left(iS[A] + i\int \mathrm{d}^4x\, J_\mu A^\mu\right).$$

Because the Abelian Faddeev–Popov determinant is field-independent in this gauge, ghosts decouple from ordinary QED in covariant gauges.

The photon propagator in momentum space is

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 + i\epsilon}\right).$$

This is the same propagator introduced in Chapter 5, now derived as the inverse of the gauge-fixed quadratic operator.

13.7 Gauge-Fixed Yang–Mills Path Integral

For a non-Abelian gauge field A^a_μ, the field strength is

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu.$$

The Yang–Mills action is

$$S_{\mathrm{YM}} = -\frac{1}{4}\int \mathrm{d}^4x\, F^a_{\mu\nu} F^{a\mu\nu}.$$

A covariant gauge-fixing term is

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi}(\partial_\mu A^{a\mu})^2.$$

The Faddeev–Popov ghost Lagrangian is

$$\mathcal{L}_{\mathrm{gh}} = \bar{c}^a\, \partial^\mu D^{ab}_\mu c^b,$$

where

$$D^{ab}_\mu = \delta^{ab}\partial_\mu + g f^{acb} A^c_\mu$$

is the covariant derivative in the adjoint representation.

The full gauge-fixed path integral is schematically

$$Z[J] = \int \mathcal{D}A\mathcal{D}\bar{c}\mathcal{D}c\, e^{i\int \mathrm{d}^4x(\mathcal{L}_{\mathrm{YM}}+\mathcal{L}_{\mathrm{gf}}+\mathcal{L}_{\mathrm{gh}}+J\cdot A)}.$$

Unlike QED, the ghosts do not decouple. They interact with the gauge field.

13.8 First Introduction to BRST Symmetry

Gauge fixing seems to break gauge invariance because a particular gauge condition has been chosen. However, the gauge-fixed theory has a remnant symmetry called BRST symmetry. It mixes gauge fields, ghosts, and auxiliary fields.

At this stage, the details are less important than the role of the symmetry. BRST symmetry keeps track of the original gauge redundancy after gauge fixing. It ensures that physical observables are gauge independent and that unphysical states decouple from the physical spectrum.

The physical state space can be characterized in terms of BRST cohomology: physical states are BRST-closed, and states differing by BRST-exact terms are identified. This language will become more important in the later chapter on BRST symmetry.

13.9 Gauge Independence of Physical Observables

Intermediate quantities can depend on the gauge parameter ξ. Propagators, individual diagrams, and off-shell Green functions may change when ξ changes. Physical observables should not.

Examples of physical observables include S-matrix elements between physical states, cross sections, decay rates, and expectation values of gauge-invariant operators. Gauge independence of these quantities is a consistency requirement of the theory.

In perturbation theory, gauge independence may involve cancellations among many diagrams, including ghost diagrams. Therefore it is generally wrong to assign physical meaning to a single gauge-dependent diagram in isolation.

13.10 Meaning of Gauge Fixing in Quantum Theory

Gauge fixing is not the selection of a physically preferred gauge. It is a method for coordinatizing the space of physically distinct configurations. Just as one may choose coordinates on an ordinary manifold without changing the geometry, one may choose a gauge without changing physical predictions.

The path integral makes this especially clear. The original integral over A_μ includes redundant gauge copies. Gauge fixing removes the redundancy and supplies the correct determinant. Different gauges are different ways of performing the same physical integral.

This point is crucial for non-Abelian theories. Without gauge fixing and ghosts, the perturbative expansion would not correctly represent the quantum gauge system.

Summary

Gauge fields cannot be integrated naively because gauge-equivalent configurations are overcounted. Gauge fixing inserts a condition selecting one representative per gauge orbit, together with the Faddeev–Popov determinant. In Abelian covariant gauges the determinant decouples, while in non-Abelian gauge theory it becomes a ghost-field action. Ghosts are not physical particles; they are the functional-integral mechanism that preserves consistency of covariant quantization. BRST symmetry records the original gauge redundancy after gauge fixing.

13.11 Chapter Checklist

After reading this chapter, one should be able to:

- explain why the naive gauge-field path integral overcounts configurations;
- describe the role of a gauge-fixing condition;
- state the basic Faddeev–Popov identity;
- explain why determinants can be represented by ghost fields;
- distinguish ghosts from physical particles;
- write the gauge-fixed photon path integral;
- write the schematic gauge-fixed Yang–Mills path integral;
- explain why ghosts decouple in Abelian theory but not in non-Abelian theory;
- state the conceptual role of BRST symmetry;
- explain why physical observables must be gauge independent.

Chapter 14

ϕ^4 Theory as the First Interacting QFT

Remark

The previous chapters built the general machinery: canonical quantization, propagators, perturbation theory, scattering, and path integrals. The theory with interaction $\lambda\phi^4/4!$ is the simplest laboratory where all of these ideas become concrete.

14.1 Why Start with ϕ^4 Theory?

The free scalar field is exactly solvable because it is a collection of independent oscillators. To study genuine quantum field theory, we need an interaction. The simplest Lorentz-invariant self-interaction of a real scalar field in four dimensions is ϕ^4.

The theory is simple enough to compute with, but rich enough to display the main features of perturbative QFT:

- interaction vertices;
- scattering amplitudes;
- loop corrections;
- ultraviolet divergences;
- counterterms;
- renormalized masses and couplings.

For this reason, ϕ^4 theory is the standard first interacting quantum field theory.

14.2 The Lagrangian

The classical Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

The free part is

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2,$$

and the interaction part is

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

The factor $4!$ is conventional. It cancels the number of permutations of four identical scalar fields at a vertex, giving a simple Feynman rule.

The classical equation of motion is

$$(\Box + m^2)\phi + \frac{\lambda}{3!}\phi^3 = 0.$$

Unlike the free Klein–Gordon equation, this equation is nonlinear. The nonlinearity is the classical shadow of particle interactions in the quantum theory.

14.3 Dimensional Analysis

In four spacetime dimensions, the action is dimensionless in units $\hbar = c = 1$. Since

$$S = \int \mathrm{d}^4x\, \mathcal{L},$$

the Lagrangian density has mass dimension four. From the kinetic term one finds

$$[\phi] = 1.$$

Therefore

$$[\lambda] = 0.$$

The coupling λ is dimensionless in four dimensions. This is one reason ϕ^4 theory is important: it is power-counting renormalizable in four dimensions.

In dimensions other than four, the mass dimension of λ changes. This will become important in the discussion of renormalization and effective field theory.

14.4 Path-Integral Generating Functional

The generating functional is

$$Z[J] = \int \mathcal{D}\phi \, \exp\left(i \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 + J\phi\right]\right).$$

It can be written in terms of the free generating functional as

$$Z[J] = \exp\left[-i\frac{\lambda}{4!}\int \mathrm{d}^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)^4\right] Z_0[J].$$

Expanding the exponential in powers of λ generates perturbation theory. The free functional $Z_0[J]$ supplies propagators; the derivative operator supplies interaction vertices.

This is the functional-integral version of the Dyson expansion and Wick's theorem.

14.5 Feynman Rules in Position Space

The basic ingredients are simple. The free propagator is

$$D_F(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

Each interaction vertex contributes

$$-i\lambda \int \mathrm{d}^4x.$$

Four scalar lines meet at each vertex. Correlation functions are obtained by summing over all contractions, or equivalently all diagrams, with the appropriate symmetry factors.

Position-space diagrams are useful conceptually because they show that each vertex is integrated over spacetime. Momentum-space diagrams are usually more convenient for scattering amplitudes because translation invariance turns those integrals into momentum-conserving delta functions.

14.6 Feynman Rules in Momentum Space

In momentum space, the propagator is

$$\frac{i}{k^2 - m^2 + i\epsilon}.$$

The four-point vertex contributes

$$-i\lambda.$$

At each vertex, four-momentum is conserved. For each undetermined internal momentum, one integrates

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}.$$

Finally, one includes the symmetry factor of the diagram.

These rules compute contributions to correlation functions and, after LSZ reduction, to scattering amplitudes.

14.7 Tree-Level $2 \to 2$ Scattering

The simplest nontrivial process is

$$\phi\phi \longrightarrow \phi\phi.$$

At lowest order there is one four-point vertex. The amplitude is

$$i\mathcal{M}_{\text{tree}} = -i\lambda, \qquad \mathcal{M}_{\text{tree}} = -\lambda.$$

This is a contact interaction. There is no internal propagator at this order.

The full matrix element contains the momentum-conserving delta function:

$$\langle p_3, p_4 | i\hat{T} | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4)\mathcal{M}_{\text{tree}}.$$

This is the same structure introduced in scattering theory, now obtained from a specific interacting Lagrangian.

14.8 First Loop Correction: The Tadpole

The two-point function receives a one-loop correction from the tadpole diagram. In momentum space, the loop integral has the form

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\epsilon}.$$

This integral diverges at large $|\ell|$. The divergence is ultraviolet: it comes from arbitrarily short-distance fluctuations.

The tadpole correction shifts the mass parameter. This is the first appearance of an important lesson: the parameters in the Lagrangian are not automatically the physical measured parameters. Quantum fluctuations correct them.

14.9 One-Loop Four-Point Correction

The four-point function receives one-loop corrections from diagrams with two vertices and two internal lines. A typical loop integral has the schematic form

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\epsilon} \frac{i}{(\ell + p)^2 - m^2 + i\epsilon}.$$

There are contributions in the s-, t-, and u-channels, corresponding to different ways of routing the external momenta through the loop.

These loop corrections modify the effective strength of the four-point interaction. In other words, they correct the coupling λ.

14.10 Ultraviolet Divergences

The loop integrals of ϕ^4 theory are typically divergent. This does not mean the theory is useless. It means that the parameters appearing in the Lagrangian must be related carefully to measured quantities.

A regulator is introduced to make integrals finite. Examples include a momentum cutoff Λ or dimensional regularization. A regulated divergent integral might depend on Λ, or contain poles in $4 - d$ when computed in d dimensions.

The goal is not to keep the regulator as a physical object. The goal is to absorb regulator dependence into redefinitions of parameters and then express predictions in terms of finite measured quantities.

14.11 Counterterms

To renormalize the theory, write the Lagrangian in terms of renormalized parameters plus counterterms:

$$\mathcal{L} = \frac{1}{2} Z_\phi \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} Z_m m^2 \phi^2 - \frac{Z_\lambda \lambda}{4!} \phi^4.$$

Equivalently, one may write

$$\mathcal{L} = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}},$$

where

$$\mathcal{L}_{\text{ct}} = \frac{1}{2} \delta Z \, \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} \delta m^2 \phi^2 - \frac{\delta \lambda}{4!} \phi^4.$$

The counterterms are chosen to cancel the divergences from loop diagrams.

In ϕ^4 theory in four dimensions, the required counterterms have the same form as terms already present in the Lagrangian. This is the practical meaning of perturbative renormalizability.

14.12 Renormalized Parameters

The mass and coupling in the Lagrangian are not directly observable until a renormalization prescription is chosen. A renormalization condition defines the parameters in terms of physical or reference quantities.

For example, one may define the physical mass by the pole of the full propagator. The renormalized coupling may be defined by the value of the four-point amplitude at a chosen reference momentum scale.

Different schemes define parameters differently, but physical predictions agree when expressed in terms of measured quantities. The scheme dependence of intermediate parameters is the beginning of the renormalization group story.

14.13 Connected and One-Particle-Irreducible Functions

The generating functional $Z[J]$ produces all diagrams. The functional $W[J] = -i \ln Z[J]$ produces connected diagrams. The effective action $\Gamma[\phi_c]$ produces one-particle-irreducible diagrams.

In ϕ^4 theory:

- the connected two-point function gives the full propagator;
- the 1PI two-point function gives the self-energy;
- the 1PI four-point function gives the quantum-corrected vertex;
- counterterms are fixed by imposing conditions on these functions.

This is the language used systematically in renormalization.

14.14 Perturbative Expansion and Its Meaning

A typical observable is computed as a formal power series:

$$\mathcal{M} = \mathcal{M}^{(0)} + \lambda \mathcal{M}^{(1)} + \lambda^2 \mathcal{M}^{(2)} + \cdots .$$

The superscript does not always count loops directly, but higher powers of λ generally correspond to more vertices and more complicated diagrams.

Perturbation theory is useful when λ is small enough that the first few terms give a good approximation. It is not a complete nonperturbative definition of the theory. Nevertheless, it is the main calculational tool for weakly coupled QFT.

14.15 Physical Lessons from ϕ^4 Theory

The theory teaches several lessons that recur throughout QFT:

- interactions are represented by vertices;
- amplitudes are built from propagators and vertices;
- loop diagrams express virtual quantum fluctuations;
- loop integrals can diverge at high momentum;
- counterterms absorb divergences into parameter definitions;
- measured masses and couplings are renormalized quantities.

These ideas reappear in QED, non-Abelian gauge theory, the Standard Model, and effective field theory.

Summary

The ϕ^4 model is the first complete interacting QFT laboratory. Its single four-scalar vertex gives tree-level scattering. Its loop diagrams produce ultraviolet divergences that require regularization and counterterms. The theory introduces the distinction between bare and renormalized parameters and shows why physical predictions must be expressed in terms of measured quantities rather than unrenormalized Lagrangian parameters.

14.16 Chapter Checklist

After reading this chapter, one should be able to:

- write the ϕ^4 Lagrangian and identify free and interacting parts;
- explain why the factor 4! is included;

- derive the basic propagator and vertex Feynman rules;
- compute the tree-level $2 \to 2$ amplitude;
- identify the tadpole as a one-loop mass correction;
- describe the one-loop correction to the four-point function;
- explain what an ultraviolet divergence is;
- state why counterterms are introduced;
- distinguish bare and renormalized parameters;
- explain why ϕ^4 theory is the first useful laboratory for renormalization.

Chapter 15

Yukawa Theory

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

15.1 Planned Role

This chapter will study a scalar field coupled to fermions, Yukawa vertices, simple scattering and decay processes, and the low-energy interpretation of scalar exchange.

Chapter 16

Quantum Electrodynamics

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

16.1 Planned Role

This chapter will introduce QED as the quantum theory of charged matter and light, including Feynman rules, spinor chains, Ward identities, and standard tree-level processes.

Chapter 17

Loop Effects in QED

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

17.1 Planned Role

This chapter will cover radiative corrections in QED, including self-energies, vacuum polarization, vertex corrections, charge renormalization, infrared divergences, and inclusive observables.

Chapter 18

Regularization

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

18.1 Planned Role

This chapter will explain why divergent loop integrals appear and how cutoff, Pauli–Villars, dimensional regularization, and subtraction schemes organize intermediate infinities.

Chapter 19

Renormalization

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

19.1 Planned Role

This chapter will develop the standard QFT treatment of renormalized masses, fields, couplings, counterterms, and physical predictions.

Chapter 20

Renormalization Group

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

20.1 Planned Role

This chapter will organize the scale dependence of quantum field theories and introduce running couplings, beta functions, fixed points, and effective descriptions.

Chapter 21

Effective Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

21.1 Planned Role

This chapter will introduce effective field theory, separation of scales, matching, power counting, symmetries, operator bases, decoupling, and naturalness.

Chapter 22

Non-Abelian Gauge Theory as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

22.1 Planned Role

This chapter will develop quantum non-Abelian gauge fields, self-interactions, gauge-fixed Feynman rules, ghosts, Ward-type identities, and unitarity checks.

Chapter 23

BRST Symmetry

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

23.1 Planned Role

This chapter will introduce BRST transformations, ghost number, nilpotency, physical states as cohomology, gauge independence, and Slavnov–Taylor identities.

Chapter 24

Quantum Chromodynamics

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

24.1 Planned Role

This chapter will introduce QCD field content, color charge, QCD Feynman rules, gluon self-interactions, asymptotic freedom, confinement, jets, and perturbative versus nonperturbative regimes.

Chapter 25

Symmetries in Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

25.1 Planned Role

This chapter will translate classical symmetries into quantum operators, conserved charges, Ward identities, current correlation functions, selection rules, and constraints on observables.

Chapter 26

Anomalies

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

26.1 Planned Role

This chapter will study symmetries that fail after quantization, chiral currents, triangle diagrams, path-integral measure effects, gauge anomalies, anomaly cancellation, and physical consequences.

Chapter 27

Vacuum Structure and Nonperturbative Effects

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

27.1 Planned Role

This chapter will discuss the quantum vacuum, degenerate vacua, tunneling, instantons, theta vacua, false vacuum decay, semiclassical approximation, and nonperturbative contributions.

Chapter 28

The Quantum Electroweak Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

28.1 Planned Role

This chapter will introduce the quantum electroweak sector, gauge fixing after symmetry breaking, massive vector boson propagators, currents, weak processes, and the low-energy Fermi theory.

Chapter 29

The Standard Model Lagrangian as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

29.1 Planned Role

This chapter will organize the Standard Model field content, representations, gauge interactions, Yukawa sector, Higgs sector, flavor mixing, neutrino masses, anomaly cancellation, and EFT viewpoint.

Chapter 30

Selected Standard Model Processes

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

30.1 Planned Role

This chapter will connect QFT calculations with representative Standard Model predictions and collider observables.

Chapter 31

Euclidean Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

31.1 Planned Role

This chapter will introduce Wick rotation, Euclidean correlation functions, reflection positivity, statistical field theory connections, finite temperature, boundary conditions, and Matsubara frequencies.

Chapter 32

Lattice Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

32.1 Planned Role

This chapter will introduce lattice regularization, scalar fields on a lattice, gauge fields on links, Wilson action, Wilson loops, fermion difficulties, Monte Carlo sampling, and the continuum limit.

Chapter 33

Large-N and Semiclassical Methods

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

33.1 Planned Role

This chapter will introduce large-N counting, planar diagrams, saddle-point methods, semiclassical expansion, collective fields, solitons, instanton calculus, and reliability of approximations.

Chapter 34

Conceptual Synthesis I

34.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 35

Conceptual Synthesis II

35.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 36

Summary: The Architecture of Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

36.1 Planned Role

This chapter will summarize quantization, Fock space, propagators, interactions, renormalization, gauge theory, anomalies, nonperturbative effects, the Standard Model, and open directions.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \quad \longrightarrow \quad \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \qquad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\mathrm{PB}} \quad \longrightarrow \quad \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word "distribution" is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green's Function to Propagator

A classical Green's function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle.$$

It is related to Green's functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

Appendix B

Reference B

Placeholder appendix.

Appendix C

Reference C

Placeholder appendix.

Appendix D

Reference D

Placeholder appendix.

Appendix E

Reference E

Placeholder appendix.

List of Figures